OAVS TGT COMPUTER TEACHER EXAMINATION

MCQ QUESTION BANK

Chapter 8 — Integrity Constraints

8.1 Entity • Domain • Referential Integrity Rules
Assertion Concept • Trigger Concept

300 Questions | 11 Sections | Easy / Moderate / Difficult Mix

Strictly syllabus-bound • Balanced answer-key distribution • Exam-oriented explanations

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SECTION 1

Introduction to Integrity Constraints

Q1 | **EASY** | **Introduction**

What is the primary purpose of integrity constraints in a database?

A. To maintain accuracy and consistency of data

B. To encrypt sensitive data

C. To increase the storage size of tables

D. To speed up query execution only

Ans: A

Explanation

Integrity constraints exist to ensure that the data stored in a database remains accurate, valid, and consistent at all times.

Q2 | **EASY** | **Introduction**

Which of the following is NOT one of the integrity rules discussed under Entity-Domain-Referential integrity?

A. Domain Integrity

B. Referential Integrity

C. Normalization Integrity

D. Entity Integrity

Ans: C

Explanation

“Normalization Integrity” is not a recognized integrity rule; normalization is a separate design process, not an integrity constraint.

Q3 | **EASY** | **Introduction**

Who is responsible for enforcing integrity constraints in a relational database?

A. The network administrator

B. The application's front-end only

C. The end user manually

D. The Database Management System (DBMS)

Ans: D

Explanation

The DBMS enforces integrity constraints automatically whenever data is inserted, updated, or deleted.

Q4 | **MODERATE** | **Introduction**

Which two mechanisms allow a database to enforce conditions that go beyond simple key-based rules?

A. Entity integrity and domain integrity

B. Normal forms and functional dependencies

C. Primary key and foreign key

D. Assertions and triggers

Ans: D

Explanation

Assertions and triggers allow enforcement of complex, condition-based or event-based rules beyond basic entity/domain/referential integrity.

Q5 | **MODERATE** | **Introduction**

What is the most likely consequence if integrity constraints are removed from a database?

A. Data may become inconsistent or invalid

B. Storage space will automatically reduce

C. Queries will run faster with no side effects

D. Tables will be automatically normalized

Ans: A

Explanation

Without integrity constraints, invalid, duplicate, or inconsistent data can be inserted, compromising the reliability of the database.

Q6 | **EASY** | **Introduction**

Entity integrity and referential integrity are both concerned with which key type respectively?

A. Composite key and unique key

B. Candidate key and super key

C. Primary key and foreign key

D. Foreign key and primary key

Ans: C

Explanation

Entity integrity is concerned with the primary key, while referential integrity governs the relationship involving the foreign key.

Q7 | **MODERATE** | **Introduction**

Which of the following best describes domain integrity?

A. Rule ensuring foreign keys match primary keys

B. Rule ensuring no duplicate rows exist

C. Rule ensuring attribute values come from a valid, defined set

D. Rule ensuring tables are in third normal form

Ans: C

Explanation

Domain integrity ensures that every value stored in a column belongs to the valid domain (data type, range, or allowed set) defined for that attribute.

Q8 | **DIFFICULT** | **Introduction**

Which statement correctly distinguishes a constraint from an assertion?

A. A constraint typically applies to a single table/column; an assertion can express a condition over the entire database

B. Assertions can only be applied to primary keys

C. Constraints and assertions are identical in every respect

D. A constraint applies to the whole database; an assertion applies to one column

Ans: A

Explanation

Constraints (entity, domain, referential) usually apply at the table or column level, whereas an assertion can enforce a condition spanning multiple tables or the whole database.

SECTION 2

Entity Integrity Rules

Q9 | **EASY** | **Entity Integrity**

Entity integrity is primarily concerned with which component of a relation?

A. Composite index

B. Foreign key

C. Candidate attribute

D. Primary key

Ans: D

Explanation

Entity integrity is defined specifically in terms of the primary key of a relation.

Q10 | **EASY** | **Entity Integrity**

According to the entity integrity rule, the primary key of a table:

A. Can be duplicated across rows

B. May contain NULL in one column only

C. Must always be a composite key

D. Cannot contain NULL values

Ans: D

Explanation

Entity integrity requires that no attribute participating in the primary key may contain a NULL value.

Q11 | **EASY** | **Entity Integrity**

What is the main purpose of entity integrity?

A. To uniquely identify every row/entity in a table

B. To ensure foreign key values match a parent table

C. To restrict the data type of an attribute

D. To automatically execute actions on data change

Ans: A

Explanation

Entity integrity ensures every row (entity) in a table can be uniquely and unambiguously identified through its primary key.

Q12 | **MODERATE** | **Entity Integrity**

If a table's primary key is composite (made of two columns), entity integrity requires that:

A. Only one column needs to be unique

B. At least one of the two columns may be NULL

C. Both columns must be foreign keys

D. Both columns together must be unique and neither may be NULL

Ans: D

Explanation

In a composite primary key, no individual attribute of the key may be NULL, and the combination of values must be unique.

Q13 | **EASY** | **Entity Integrity**

Which of the following would violate entity integrity in a STUDENT table with RollNo as primary key?

A. A student's marks exceeding 100 due to a typing error

B. A NULL value in the RollNo column

C. A student having no phone number

D. Two students having different names

Ans: B

Explanation

A NULL primary key value directly violates entity integrity, since the primary key must always uniquely identify a row.

Q14 | **MODERATE** | **Entity Integrity**

Two rows in the EMPLOYEE table both have EmpID = 101. What does this indicate?

A. A valid situation, since EmpID is not unique

B. An entity integrity violation

C. A referential integrity violation

D. A domain integrity violation

Ans: B

Explanation

Duplicate primary key values break the uniqueness requirement of entity integrity, which mandates that every primary key value be unique.

Q15 | **EASY** | **Entity Integrity**

Why can't a primary key value be NULL?

A. Because NULL takes more storage space

B. Because primary keys must always be numeric

C. Because NULL is not a valid data type

D. Because a NULL value cannot uniquely identify a row

Ans: D

Explanation

A NULL value represents 'unknown' or 'missing', which cannot serve to uniquely identify a specific row, defeating the purpose of a primary key.

Q16 | **MODERATE** | **Entity Integrity**

Which of these is an example of a table correctly satisfying entity integrity?

A. A table with no primary key defined at all

B. A table having two rows with identical PK values

C. A table where PK column has one NULL entry among ten rows

D. A table where every PK value is unique and none is NULL

Ans: D

Explanation

Entity integrity is satisfied only when all primary key values are both unique and non-NULL.

Q17 | **EASY** | **Entity Integrity**

Entity integrity applies to which type of key exclusively?

A. Only the chosen primary key

B. Only unique but non-key attributes

C. Only foreign keys

D. Any candidate key of the table

Ans: A

Explanation

While a table may have several candidate keys, entity integrity specifically governs the one selected as the primary key.

Q18 | **DIFFICULT** | **Entity Integrity**

A table has a candidate key that is NOT chosen as the primary key, and this candidate key has a NULL value in one row. Does this violate entity integrity?

A. Yes, because all candidate keys must obey entity integrity

B. No, because entity integrity applies only to the declared primary key

C. No, because candidate keys are never enforced by the DBMS

D. Yes, because NULLs are never allowed anywhere in a table

Ans: B

Explanation

Entity integrity is defined strictly for the primary key; a non-primary candidate key is not bound by this rule unless separately constrained as NOT NULL/UNIQUE.

Q19 | **MODERATE** | **Entity Integrity**

In a table with StudentID as the primary key, which insertion would violate entity integrity?

A. Inserting a student with a duplicate email address

B. Inserting a student with an unusually long name

C. Inserting a student record with StudentID left blank/NULL

D. Inserting a new student with a new unique StudentID

Ans: C

Explanation

Leaving the primary key column blank (NULL) directly breaks the entity integrity rule.

Q20 | **EASY** | **Entity Integrity**

Which term best describes what entity integrity guarantees about each row of a table?

A. Referential correctness

B. Unique identifiability

C. Format correctness

D. Data type correctness

Ans: B

Explanation

Entity integrity guarantees that every row can be uniquely identified via a non-NULL, unique primary key.

Q21 | **MODERATE** | **Entity Integrity**

A BOOK table uses ISBN as its primary key. Which scenario represents a valid entity-integrity-compliant insertion?

A. A new book with a completely new, unique ISBN

B. A new book with ISBN containing only spaces

C. A new book inserted with ISBN left empty

D. A new book with an ISBN identical to an existing book

Ans: A

Explanation

A unique, non-NULL ISBN satisfies the entity integrity requirement for the primary key.

Q22 | **DIFFICULT** | **Entity Integrity**

Which of the following is a common misconception about entity integrity?

A. That it ensures rows are uniquely identifiable

B. That any column marked UNIQUE (even non-key) automatically enforces entity integrity

C. That NULL primary keys are disallowed

D. That it applies only to the primary key

Ans: B

Explanation

A common misconception is assuming any UNIQUE column enforces entity integrity; in fact, entity integrity is formally tied only to the primary key, not to every unique column.

Q23 | **EASY** | **Entity Integrity**

Fill in the blank: Entity integrity ensures that the primary key value is always _____ and _____.

A. numeric; short

B. encrypted; hidden

C. unique; non-NULL

D. duplicated; optional

Ans: C

Explanation

The two defining requirements of entity integrity are uniqueness and non-NULL values for the primary key.

Q24 | **MODERATE** | **Entity Integrity**

Which real-world example best illustrates entity integrity?

A. Every student is assigned a unique, compulsory Roll Number

B. A student cannot enroll in a course that does not exist

C. An automatic email is sent when a student registers

D. A course fee must be a positive number

Ans: A

Explanation

Assigning every student a unique, compulsory (non-NULL) Roll Number is a direct real-world application of entity integrity via the primary key.

Q25 | **EASY** | **Entity Integrity**

A relation without any primary key defined:

A. Is a violation of domain integrity

B. Automatically satisfies entity integrity

C. Cannot have entity integrity enforced meaningfully

D. Is a violation of referential integrity

Ans: C

Explanation

Since entity integrity is defined in terms of the primary key, a relation with no primary key has no basis for this rule to be meaningfully enforced.

Q26 | **MODERATE** | **Entity Integrity**

Which of the following pairs correctly matches the entity integrity rule with its restriction?

A. Foreign key -> must reference itself

B. Primary key -> must be unique and non-NULL

C. Primary key -> may repeat

D. Foreign key -> must always be NULL

Ans: B

Explanation

The entity integrity rule specifically requires the primary key to be unique and non-NULL.

Q27 | **DIFFICULT** | **Entity Integrity**

Assertion: A table can never have two identical rows if entity integrity is enforced.
Reason: Entity integrity guarantees a unique, non-NULL primary key for every row.
Choose the correct option.

A. Both Assertion and Reason are true, and Reason correctly explains Assertion

B. Assertion is false, Reason is true

C. Both are true, but Reason does not explain Assertion

D. Assertion is true, Reason is false

Ans: A

Explanation
Since every row must have a unique, non-NULL primary key, no two rows can be identical in their key values, which directly explains why full-row duplication of identity is prevented.

Q28 | **MODERATE** | **Entity Integrity**

In a PATIENT table, PatientID is the primary key. A data-entry clerk tries to insert a new patient without providing a PatientID. What will the DBMS typically do if entity integrity is enforced?

A. Silently convert PatientID to zero

B. Reject the insertion

C. Accept the row and auto-generate a blank key

D. Accept it but mark the row as inactive

Ans: B

Explanation
A DBMS enforcing entity integrity rejects any insertion that would leave the primary key NULL.

Q29 | **EASY** | **Entity Integrity**

Which of the following is TRUE regarding entity integrity?

A. It only applies to numeric primary keys

B. It is unrelated to the concept of a primary key

C. It ensures each row is distinguishable from every other row

D. It allows duplicate primary keys if the table is large

Ans: C

Explanation
Entity integrity's core purpose is to make sure each row can be distinguished from every other row via a unique, non-NULL primary key.

Q30 | **MODERATE** | **Entity Integrity**

A LIBRARY table has BookCode as primary key. Which update would violate entity integrity?

A. Changing the number of available copies

B. Changing the publisher name

C. Changing BookCode of one row to match another existing row's BookCode

D. Changing a book's title

Ans: C

Explanation

Updating a primary key value so that it duplicates another row's key violates the uniqueness requirement of entity integrity.

Q31 | **EASY** | **Entity Integrity**

Entity integrity is one of the rules under which broader category of database rules?

A. Integrity constraints

B. Query optimization rules

C. Indexing strategies

D. Normalization rules

Ans: A

Explanation

Entity integrity, along with domain and referential integrity, falls under the broader category of integrity constraints.

Q32 | **DIFFICULT** | **Entity Integrity**

Which scenario is a subtle but real violation of entity integrity, even though no error message may appear in a poorly designed system?

A. A table with an auto-incrementing numeric ID

B. A table with a foreign key referencing another table

C. A table with a column restricted to values between 1 and 100

D. A table where the 'primary key' column is not actually enforced as NOT NULL/UNIQUE by the DBMS, allowing duplicate or NULL values

Ans: D

Explanation

If a DBMS is not properly configured to enforce NOT NULL and UNIQUE on the primary key, entity integrity can be silently violated despite the column being labeled as the primary key.

Q33 | **MODERATE** | **Entity Integrity**

Which of the following best explains why entity integrity is essential for data retrieval?

A. It speeds up disk read operations

B. It compresses the table for faster backups

C. It removes the need for indexes

D. It allows unambiguous identification and retrieval of a specific row

Ans: D

Explanation

Because the primary key uniquely identifies each row, entity integrity allows the DBMS and users to retrieve or reference an exact, unambiguous row.

Q34 | **EASY** | **Entity Integrity**

Which of the following column properties are jointly required by entity integrity?

A. Foreign key and CHECK

B. NULL allowed and UNIQUE

C. NOT NULL and UNIQUE

D. NOT NULL and duplicate-allowed

Ans: C

Explanation

Entity integrity requires the primary key to be both NOT NULL and UNIQUE.

Q35 | **MODERATE** | **Entity Integrity**

Match the following:
(i) Primary key with NULL value
(ii) Primary key duplicated in two rows
(a) Violates uniqueness
(b) Violates non-NULL rule

A. (i)-(b), (ii)-(a)

B. (i)-(b), (ii)-(b)

C. (i)-(a), (ii)-(a)

D. (i)-(a), (ii)-(b)

Ans: A

Explanation

A NULL primary key violates the non-NULL rule (b), while a duplicated primary key violates the uniqueness rule (a).

Q36 | **MODERATE** | **Entity Integrity**

ENROLLMENT has the composite primary key (RollNo, SubjectCode) and currently holds the rows (1, M1), (1, M2) and (2, M1). Which new row violates entity integrity?

A. (1, M3)

B. (1, M1)

C. (2, M2)

D. (3, M1)

Ans: B

Explanation

The combination (1, M1) already exists, so the composite key would be duplicated. Each other row forms a new unique combination.

Q37 | **MODERATE** | **Entity Integrity**

In a table with the composite primary key (OrderID, ItemNo), a row (501, NULL) is inserted. What is the outcome under entity integrity?

A. It is rejected, because no attribute of a composite primary key may be NULL

B. It is accepted only if OrderID is unique in the table

C. It is accepted, because the pair is still unique

D. It is accepted, because OrderID alone is non-NULL

Ans: A

Explanation

Entity integrity forbids NULL in every attribute that forms the primary key, even when the other part is filled in.

Q38 | **DIFFICULT** | **Entity Integrity**

StudentID is the numeric primary key of a table. One row has StudentID = 0 and no other row has 0. Does this violate entity integrity?

A. Yes, because a primary key must be greater than zero

B. Yes, because 0 is equivalent to NULL

C. No, but only if StudentID is also a foreign key

D. No, because 0 is an actual value, not NULL, and it is unique

Ans: D

Explanation

NULL means 'unknown or missing', which is different from the number 0. A unique, non-NULL 0 is a valid primary key value.

Q39 | **EASY** | **Entity Integrity**

Which update operation on EMPLOYEE (primary key EmpID) violates entity integrity?

A. Changing the Name of one employee

B. Changing the EmpID of one employee to NULL

C. Changing the Salary of every employee to the same value

D. Changing the Salary of one employee to a higher value

Ans: B

Explanation

Setting the primary key to NULL breaks the non-NULL rule. Changing non-key attributes does not affect entity integrity.

Q40 | **DIFFICULT** | **Entity Integrity**

Which operation can never, by itself, cause a violation of entity integrity?

A. Deleting a row from the table

B. Inserting a row with a NULL primary key

C. Inserting a row with an existing primary key value

D. Updating a primary key to a value that already exists

Ans: A

Explanation

Deleting a row only removes a key value and cannot create a NULL or duplicate key. Only INSERT and UPDATE of the key can violate entity integrity.

Q41 | **MODERATE** | **Entity Integrity**

Consider the statements:
I. Every primary key value must be unique.
II. A primary key column may hold one NULL value.
III. Entity integrity is defined for the primary key.
Which statements are correct?

A. I and II only

B. I and III only

C. I, II and III

D. II and III only

Ans: B

Explanation

A primary key can never be NULL, not even once, so II is wrong. I and III are correct.

Q42 | **DIFFICULT** | **Entity Integrity**

Foreign keys in other tables refer to a table's primary key. Which rule must hold in the referenced table so that each reference points to exactly one row?

A. Domain integrity: every attribute has a data type

B. Assertion: the table must contain at least one row

C. Referential integrity: the primary key must be a foreign key

D. Entity integrity: the primary key is unique and non-NULL

Ans: D

Explanation

A reference identifies one parent row only if the primary key values are unique and non-NULL, which is what entity integrity guarantees.

Q43 | **MODERATE** | **Entity Integrity**

Assertion (A): A primary key column cannot contain NULL.
Reason (R): A NULL value means the value is unknown, so it cannot identify a row.
Choose the correct option.

A. Both A and R are true, but R does not explain A

B. A is false but R is true

C. A is true but R is false

D. Both A and R are true, and R explains A

Ans: D

Explanation

NULL stands for an unknown value, which cannot identify a row. That is exactly why entity integrity bans it in the primary key.

Q44 | **MODERATE** | **Entity Integrity**

A bulk upload of 500 student records contains two rows with the same AdmissionNo (the primary key). What will a DBMS enforcing entity integrity do with the offending row?

A. Reject the row that would duplicate the key

B. Accept both rows but flag them as duplicates

C. Accept the second row and delete the first

D. Accept both rows and merge them automatically

Ans: A

Explanation

A duplicate primary key value breaks uniqueness, so the DBMS refuses the row that would create it.

Q45 | **EASY** | **Entity Integrity**

Which of the following is NOT required by entity integrity?

A. The primary key must refer to a row in another table

B. The primary key must be unique

C. Each row must be identifiable through its primary key

D. The primary key must not be NULL

Ans: A

Explanation

Referring to another table is the job of a foreign key (referential integrity), not of entity integrity.

Q46 | **EASY** | **Entity Integrity**

Which set of values could fill a primary key column without violating entity integrity?

A. 101, 102, 102, 104

B. 101, 102, NULL, 104

C. NULL, NULL, NULL, NULL

D. 101, 102, 103, 104

Ans: D

Explanation

Only the first set is both unique and free of NULL.

Q47 | **MODERATE** | **Entity Integrity**

In EMPLOYEE (EmpID, Name, Dept) with primary key EmpID, employee 10 works in Sales and a new HR employee is also given EmpID 10. Is entity integrity violated?

A. Yes, but only if both employees have the same Name

B. No, because the two employees are in different departments

C. Yes, because primary key values must be unique across the whole table

D. No, because only Name must be unique

Ans: C

Explanation

Uniqueness of the primary key is table-wide, whatever the values in other columns.

SECTION 3

Domain Integrity Rules

Q48 | **EASY** | **Domain Integrity**

Domain integrity ensures that the values stored in a column:

A. Always reference another table

B. Are automatically encrypted

C. Belong to a defined, valid set of allowable values

D. Are always unique across the table

Ans: C

Explanation

Domain integrity restricts the values of an attribute to a valid domain, such as a specific data type, range, or set of allowed values.

Q49 | **EASY** | **Domain Integrity**

Which of the following is the main purpose of domain integrity?

A. To restrict attribute values to valid, meaningful data

B. To automatically trigger actions on updates

C. To link two related tables

D. To uniquely identify each row

Ans: A

Explanation

Domain integrity's main purpose is ensuring that column values are valid and meaningful according to a defined domain.

Q50 | **MODERATE** | **Domain Integrity**

A column 'Gender' is restricted to only 'M' or 'F'. Inserting the value 'X' into this column would violate:

A. Domain integrity

B. Referential integrity

C. Entity integrity

D. None of the above

Ans: A

Explanation

Restricting a column to a specific set of allowed values is a domain integrity rule; inserting a value outside that set violates domain integrity.

Q51 | **EASY** | **Domain Integrity**

Which of these is an example of a domain restriction?

A. A foreign key must match a primary key

B. Age must be a positive integer between 0 and 120

C. A student cannot be deleted if enrolled in a course

D. A primary key cannot be NULL

Ans: B

Explanation

Restricting Age to positive integers within a sensible range is a classic domain integrity (range) restriction.

Q52 | **MODERATE** | **Domain Integrity**

Which type of restriction is NOT typically considered part of domain integrity?

A. Allowed-value/format restriction

B. Range restriction

C. Uniqueness of the primary key

D. Data type restriction

Ans: C

Explanation

Uniqueness of the primary key is governed by entity integrity, not domain integrity.

Q53 | **EASY** | **Domain Integrity**

If the 'Marks' column is defined to accept only integer values from 0 to 100, entering the value 150 would cause:

A. No violation at all

B. A domain integrity violation

C. An entity integrity violation

D. A referential integrity violation

Ans: B

Explanation

A value outside the defined valid range (0-100) breaches the domain integrity rule for that column.

Q54 | **MODERATE** | **Domain Integrity**

A 'DateOfBirth' column is defined with data type DATE. Entering the text 'Hello' into this column would violate:

A. Domain integrity

B. Entity integrity

C. Referential integrity

D. No rule is violated

Ans: A

Explanation

Storing a non-date text value in a DATE column breaks the data-type restriction, which is part of domain integrity.

Q55 | **EASY** | **Domain Integrity**

Domain integrity is enforced mainly at which level?

A. Database level only

B. Index level

C. Row level

D. Column/attribute level

Ans: D

Explanation

Domain integrity is defined and enforced at the level of an individual attribute/column, restricting the values it may hold.

Q56 | **DIFFICULT** | **Domain Integrity**

A 'Salary' column is defined as DECIMAL and must be greater than 0. Which value would violate domain integrity?

A. 0.01

B. 25000.50

C. -5000

D. 99999.99

Ans: C

Explanation

A negative salary value (-5000) violates the defined domain restriction that salary must be greater than 0.

Q57 | **EASY** | **Domain Integrity**

Which of the following best defines a 'domain' in the context of domain integrity?

A. A type of database index

B. A network address used by the database server

C. A synonym for the primary key

D. The set of all valid values an attribute can take

Ans: D

Explanation

A domain is the set of all permissible/legal values that an attribute is allowed to hold.

Q58 | **MODERATE** | **Domain Integrity**

A column 'Email' is expected to always contain the '@' symbol. Enforcing this expectation is an example of which restriction?

A. Format/allowed-value restriction

B. Referential restriction

C. Entity restriction

D. Range restriction

Ans: A

Explanation

Ensuring a value follows a required pattern (such as containing '@') is a format-based domain restriction.

Q59 | **EASY** | **Domain Integrity**

Which of the following is a valid value for a column 'Status' restricted to the domain {'Active','Inactive'}?

A. 'Pending'

B. NULL, if not restricted otherwise

C. 'ACTIVE1'

D. 'Active'

Ans: D

Explanation

'Active' is one of the explicitly permitted values in the defined domain, so it is valid.

Q60 | **MODERATE** | **Domain Integrity**

Table EMPLOYEE has a column 'Age' with domain restricted to integers 18-60. Which insertion is INVALID?

A. Age = 17

B. Age = 59

C. Age = 45

D. Age = 25

Ans: A

Explanation

A value of 17 falls outside the valid domain range of 18 to 60, violating domain integrity.

Q61 | **DIFFICULT** | **Domain Integrity**

Which of the following is a common misconception about domain integrity?

A. That it can never be checked using a CHECK-type constraint

B. That it is unrelated to the concept of a valid value set

C. That it can involve data type, range, and allowed-value restrictions together

D. That it only applies to numeric columns

Ans: D

Explanation

A common misconception is assuming domain integrity applies only to numeric data; in reality it applies to any attribute type, including text, dates, and categorical values.

Q62 | **EASY** | **Domain Integrity**

Fill in the blank: Domain integrity ensures that every value in a column is _____ for that attribute.

A. unique

B. valid

C. foreign

D. indexed

Ans: B

Explanation

Domain integrity's defining feature is that it ensures every stored value is valid according to the attribute's defined domain.

Q63 | **MODERATE** | **Domain Integrity**

A column 'PinCode' must always be exactly 6 digits. Entering '12345' (5 digits) would violate:

A. Referential integrity

B. Domain integrity

C. Entity integrity

D. No constraint at all

Ans: B

Explanation

A format restriction requiring exactly 6 digits is a domain integrity rule, and a 5-digit entry violates it.

Q64 | **EASY** | **Domain Integrity**

Which of the following column definitions demonstrates a domain integrity restriction?

A. FOREIGN KEY(DeptID) REFERENCES DEPT(DeptID)

B. UNIQUE(Email)

C. CHECK (Marks BETWEEN 0 AND 100)

D. PRIMARY KEY(StudentID)

Ans: C

Explanation

A CHECK constraint restricting Marks to a valid range (0-100) is a direct implementation of domain integrity.

Q65 | **MODERATE** | **Domain Integrity**

Assertion (A): Data type restriction alone is sufficient to fully enforce domain integrity.
Reason (R): Range and allowed-value restrictions are also required to catch violations that a data type check would miss.
Choose the correct option.

A. A is true and R is false

B. Both A and R are false

C. A is true and R is true

D. A is false and R is true

Ans: D

Explanation

A negative age typed as INTEGER passes a data-type check but still violates domain rules, so data type alone is insufficient (A is false); range/allowed-value checks are indeed additionally required (R is true).

Q66 | **EASY** | **Domain Integrity**

Which of the following pairs of restrictions are both commonly grouped under domain integrity?

A. Primary key and foreign key

B. Data type and range restriction

C. Insert restriction and delete restriction

D. Uniqueness and NULL restriction

Ans: B

Explanation

Data type restrictions and range restrictions are both classic examples of domain integrity rules.

Q67 | **MODERATE** | **Domain Integrity**

A table column 'BloodGroup' should only accept values from {A+, A-, B+, B-, O+, O-, AB+, AB-}. This is an example of:

A. Domain integrity using an allowed-value restriction

B. Entity integrity using a composite key

C. Assertion enforcing a database-wide rule

D. Referential integrity using an implicit foreign key

Ans: A

Explanation

Restricting a column to a fixed, explicit set of allowed values is a textbook example of domain integrity.

Q68 | **EASY** | **Domain Integrity**

Which of these violates domain integrity in a column 'Quantity' defined as a positive integer?

A. Quantity = 1

B. Quantity = 10

C. Quantity = 0.5

D. Quantity = 500

Ans: C

Explanation

A fractional value (0.5) violates the domain restriction that Quantity must be a positive integer.

Q69 | **DIFFICULT** | **Domain Integrity**

Which scenario represents a domain integrity violation that a simple data-type restriction alone would NOT catch?

A. Storing a negative value in an 'Age' column typed as INTEGER but logically restricted to be positive

B. Storing text in a numeric column

C. Storing a date in a DATE-typed column

D. Storing NULL in a nullable column

Ans: A

Explanation

A negative age is still a valid INTEGER by data type, so only an additional range/CHECK restriction (not the data type alone) can catch this domain violation.

Q70 | **MODERATE** | **Domain Integrity**

In the STUDENT table, 'Grade' is restricted to {A, B, C, D, F}. Which insertion is VALID?

A. Grade = 'B'

B. Grade = 'E'

C. Grade = 'Pass'

D. Grade = NULL, if column is NOT NULL

Ans: A

Explanation

'B' belongs to the explicitly permitted set of grade values, making it a valid entry under domain integrity.

Q71 | **EASY** | **Domain Integrity**

Domain integrity is most closely associated with which SQL-level constraint construct (conceptually)?

A. INDEX

B. CHECK

C. FOREIGN KEY

D. PRIMARY KEY

Ans: B

Explanation

The CHECK constraint is the SQL-level construct most closely associated with enforcing domain integrity rules such as ranges and allowed values.

Q72 | MODERATE | Domain Integrity

Match the following:
(i) Data type restriction
(ii) Range restriction
(iii) Allowed-value restriction
(a) Marks BETWEEN 0 AND 100
(b) Gender IN ('M','F')
(c) DOB must be of type DATE

A. (i)-(c), (ii)-(a), (iii)-(b)

B. (i)-(c), (ii)-(b), (iii)-(a)

C. (i)-(b), (ii)-(c), (iii)-(a)

D. (i)-(a), (ii)-(b), (iii)-(c)

Ans: A

Explanation
Data type restriction matches DOB as DATE (c); range restriction matches Marks between 0-100 (a); allowed-value restriction matches Gender IN ('M','F') (b).

Q73 | EASY | Domain Integrity

Which of the following best distinguishes domain integrity from entity integrity?

A. Domain integrity concerns attribute values; entity integrity concerns the primary key's uniqueness/non-NULL status

B. Domain integrity applies only to dates

C. Both are identical in every respect

D. Domain integrity concerns foreign keys; entity integrity concerns triggers

Ans: A

Explanation
Domain integrity restricts valid values for any attribute, while entity integrity is specifically about primary key uniqueness and non-NULL status.

Q74 | EASY | Domain Integrity

A column Status is defined with DEFAULT 'Active'. A row is inserted without any Status value. What happens?

A. The row is rejected

B. 'Active' is stored automatically

C. The Status of the previous row is copied

D. NULL is stored

Ans: B

Explanation
A DEFAULT clause supplies a predefined value, which belongs to the column's domain, when none is given.

Q75 | **MODERATE** | **Domain Integrity**

A column Phone is declared NOT NULL, so a row without a phone number is rejected. Which category of rule is this restriction?

A. Trigger-based integrity

B. Domain integrity, since NULL is not an allowed value

C. Referential integrity

D. Entity integrity, since Phone is the key

Ans: B

Explanation

NOT NULL removes NULL from the set of values the column may hold, which is a domain-type restriction.

Q76 | **DIFFICULT** | **Domain Integrity**

Month is stored in an INTEGER column but only the values 1 to 12 are meaningful. Which statement about domain and data type is correct?

A. The data type is narrower than the domain

B. The domain is always identical to the data type

C. A domain can be defined only for text columns

D. The domain (1 to 12) is narrower than the INTEGER data type

Ans: D

Explanation

A data type allows many values (all integers), while the domain keeps only those that are meaningful for the attribute.

Q77 | **EASY** | **Domain Integrity**

Marks is an integer column that must lie between 0 and 100. An update sets Marks to 105. Which rule is violated?

A. Referential integrity

B. Domain integrity

C. Entity integrity

D. No rule, since the data type is still integer

Ans: B

Explanation

105 is a valid integer but lies outside the allowed range, so domain integrity is violated.

Q78 | **MODERATE** | **Domain Integrity**

Consider the statements:
I. Domain integrity restricts the values an attribute can take.
II. A domain can be defined using data type, range or allowed values.
III. Domain integrity requires every value in a column to be unique.
Which statements are correct?

A. I and III only

B. I, II and III

C. I and II only

D. II and III only

Ans: C

Explanation
Uniqueness is not a requirement of domain integrity; repeated values are fine if they are valid.

Q79 | **DIFFICULT** | **Domain Integrity**

A student's age is entered as 20, but the student is really 19. The Age domain is 15 to 30. Will domain integrity detect the error?

A. No, because 20 is still a valid value within the domain

B. Yes, because 20 differs from the true age

C. Yes, because domain integrity checks factual correctness

D. No, because Age is not covered by any integrity rule

Ans: A

Explanation
Domain integrity checks that a value is allowed for the attribute, not that it is factually true.

Q80 | **MODERATE** | **Domain Integrity**

A college portal's Attendance column accepted the value 135 (percent). Which rule was missing or not enforced?

A. Entity integrity on the primary key

B. A range restriction under domain integrity

C. Referential integrity on a foreign key

D. A trigger, since no constraint can catch this

Ans: B

Explanation
A range check such as 0 to 100 is a domain rule and would have rejected 135.

Q81 | **MODERATE** | **Domain Integrity**

A Gender column holds 'M', 'F', 'Male', 'female' and 'm', which makes reports inconsistent. Which restriction would have prevented this?

A. An allowed-value restriction such as Gender IN ('M','F')

B. A longer CHAR length for Gender

C. A NOT NULL restriction alone

D. A primary key on the Gender column

Ans: A

Explanation

Listing the permitted values is an allowed-value domain restriction and keeps entries consistent.

Q82 | **MODERATE** | **Domain Integrity**

Assertion (A): A value can have the correct data type and still violate domain integrity.
Reason (R): Domain integrity also includes range and allowed-value rules.
Choose the correct option.

A. A is false but R is true

B. Both A and R are true, and R explains A

C. A is true but R is false

D. Both A and R are true, but R does not explain A

Ans: B

Explanation

Data type is only one part of a domain. A valid integer such as -5 can still break a range rule.

Q83 | **MODERATE** | **Domain Integrity**

PinCode is a CHAR(6) column that must contain only digits. Which entry fits the length but violates domain integrity?

A. 56A001

B. 110011

C. 560001

D. 700091

Ans: A

Explanation

'56A001' has six characters but contains a letter, which breaks the digits-only format rule.

Q84	DIFFICULT	Domain Integrity

Age is an INTEGER column with the rule 18 to 60. Which pair of entries is BOTH invalid?

A. NULL and 45, when NULL is allowed

B. 18 and 60

C. 17.5 and 61

D. 25 and 59

Ans: C

Explanation

17.5 is not an integer and is below 18; 61 is above 60. The other pairs are valid.

SECTION 4

Referential Integrity Rules

Q85 | **EASY** | **Referential Integrity**

Referential integrity is a rule that governs the relationship between:

A. A column and its own data type

B. Two foreign keys in different tables

C. A foreign key in one table and a primary key in another table

D. Two primary keys in the same table

Ans: C

Explanation

Referential integrity ensures that a foreign key value in one table correctly references an existing primary key value in a related table.

Q86 | **EASY** | **Referential Integrity**

What is the main purpose of referential integrity?

A. To ensure relationships between tables remain valid and consistent

B. To make sure primary keys are non-NULL

C. To restrict a column's data type

D. To automatically fire an action when data changes

Ans: A

Explanation

Referential integrity's purpose is to preserve valid, consistent relationships between related (parent-child) tables.

Q87 | **MODERATE** | **Referential Integrity**

A STUDENT table has a foreign key DeptID referencing DEPARTMENT(DeptID). Inserting a student with DeptID = 99, where no department with DeptID 99 exists, would:

A. Violate entity integrity

B. Violate domain integrity

C. Violate referential integrity

D. Cause no violation

Ans: C

Explanation

A foreign key value that does not match any existing primary key value in the referenced table is a direct violation of referential integrity.

Q88 | **EASY** | **Referential Integrity**

In referential integrity, the table containing the foreign key is commonly called the:

A. Master table

B. Reference table

C. Parent table

D. Child table

Ans: D

Explanation

The table that holds the foreign key referencing another table's primary key is called the child (or referencing) table.

Q89 | **EASY** | **Referential Integrity**

The table containing the primary key that is referenced by a foreign key is called the:

A. Parent table

B. Child table

C. Junction table

D. Detail table

Ans: A

Explanation

The table whose primary key is referenced by another table's foreign key is called the parent (or referenced) table.

Q90 | **MODERATE** | **Referential Integrity**

Which of the following is generally considered a VALID foreign key value under referential integrity (assuming NULL is allowed)?

A. A value with no match in the parent table

B. A random string unrelated to any key

C. Any negative number regardless of the parent table

D. A value matching an existing primary key value in the parent table

Ans: D

Explanation

A foreign key value is valid when it matches an existing primary key value in the parent table, or is NULL if NULLs are permitted for that column.

Q91 | **MODERATE** | **Referential Integrity**

Can a foreign key value be NULL under referential integrity?

A. Never, under any circumstance

B. Yes, if the foreign key column allows NULL, indicating the relationship is optional

C. Only if the primary key is also NULL

D. Only if the parent table is empty

Ans: B

Explanation

Unlike a primary key, a foreign key is generally allowed to be NULL (if the column permits it), representing an optional relationship.

Q92 | **EASY** | **Referential Integrity**

Which of the following actions is most likely to cause a referential integrity violation?

A. Updating a non-key column in the parent table

B. Inserting a new row in the parent table

C. Deleting a parent row that is still referenced by child rows, without any cascading rule

D. Inserting a row in the child table with a valid foreign key

Ans: C

Explanation

Deleting a referenced parent row while child rows still point to it (with no cascade or restrict rule handling it properly) breaks referential integrity.

Q93 | **MODERATE** | **Referential Integrity**

In a CUSTOMER (parent) and ORDER (child) table relationship, which update to the CUSTOMER table could break referential integrity?

A. Updating a customer's address

B. Adding a new customer

C. Deleting a customer with no existing orders

D. Changing a CustomerID value that is still referenced by existing orders, without updating those orders

Ans: D

Explanation

If a referenced primary key value is changed without correspondingly updating all child rows that reference it, those child rows become 'orphaned', violating referential integrity.

Q94 | **EASY** | **Referential Integrity**

Referential integrity primarily involves which two types of keys?

A. Composite key and surrogate key

B. Primary key and foreign key

C. Primary key and candidate key

D. Foreign key and super key

Ans: B

Explanation

Referential integrity is defined in terms of the relationship between a foreign key (in the child table) and a primary key (in the parent table).

Q95 | **MODERATE** | **Referential Integrity**

Consider DEPARTMENT(DeptID PK) and EMPLOYEE(EmpID PK, DeptID FK). Which insertion into EMPLOYEE is VALID?

A. An employee with DeptID = 500, which does not exist in DEPARTMENT

B. An employee with negative EmpID if EmpID is meant to be positive

C. An employee with DeptID that exists in DEPARTMENT

D. An employee with a duplicate EmpID

Ans: C

Explanation

A foreign key value that matches an existing DeptID in the parent DEPARTMENT table is a valid, referentially correct insertion.

Q96 | **DIFFICULT** | **Referential Integrity**

A DBMS uses ON DELETE CASCADE for a foreign key relationship. If a parent row is deleted, what happens to matching child rows?

A. The deletion of the parent row is always blocked

B. The child rows' foreign key is automatically set to a random value

C. The child rows are automatically deleted as well

D. Nothing happens to child rows

Ans: C

Explanation

ON DELETE CASCADE automatically deletes all child rows that reference the deleted parent row, preserving referential integrity.

Q97 | **MODERATE** | **Referential Integrity**

What is the basic consequence of violating referential integrity in a database?

A. Improved query performance

B. Reduced storage requirements

C. Automatic normalization of tables

D. Orphaned or inconsistent records that reference non-existent parent rows

Ans: D

Explanation

Violating referential integrity results in orphan records — child rows referencing parent rows that no longer exist or never existed.

Q98 | **EASY** | **Referential Integrity**

Which of the following best defines referential integrity?

A. A rule ensuring every attribute value is from a valid domain

B. A rule ensuring foreign key values correspond to existing primary key values in the related table

C. A rule ensuring the primary key is unique and non-NULL

D. A rule that automatically executes stored procedures

Ans: B

Explanation

Referential integrity is the rule that a foreign key value must either match an existing primary key value in the referenced table, or be NULL.

Q99 | **MODERATE** | **Referential Integrity**

In a small library database, BOOK(BookID PK) and ISSUE(IssueID PK, BookID FK). Which deletion from BOOK would create a referential integrity issue if no cascading rule exists?

A. Deleting a book that has never been issued

B. Adding a new book

C. Updating the book's title

D. Deleting a book that currently has an active issue record referencing it

Ans: D

Explanation

Deleting a BOOK row that is still referenced by an ISSUE row (without a cascade/restrict rule) leaves the ISSUE row pointing to a non-existent book, violating referential integrity.

Q100 | **DIFFICULT** | **Referential Integrity**

Which of the following best explains why referential integrity is important for maintaining relationships between tables?

A. It ensures all tables have the same number of rows

B. It prevents any table from having more than one foreign key

C. It guarantees that related data across tables stays logically connected and free of dangling references

D. It automatically merges parent and child tables

Ans: C

Explanation

Referential integrity keeps related data across multiple tables logically consistent by preventing dangling (unmatched) foreign key references.

Q101 | **EASY** | **Referential Integrity**

Fill in the blank: A foreign key value must either match a primary key value in the parent table or be _____.

A. negative

B. duplicated

C. NULL (if allowed)

D. encrypted

Ans: C

Explanation

Referential integrity permits a foreign key to be NULL (if the column allows it) as an alternative to matching a parent key value.

Q102 | **MODERATE** | **Referential Integrity**

Which scenario represents a violation of referential integrity?

A. A child table has more rows than the parent table

B. A child row's foreign key value has no corresponding row in the parent table

C. A parent row has no matching child rows

D. A parent table's primary key values are all unique

Ans: B

Explanation

When a foreign key value does not correspond to any row in the parent table, it is a classic referential integrity violation (an orphan reference).

Q103 | **MODERATE** | **Referential Integrity**

ON DELETE RESTRICT (or NO ACTION) for a foreign key means that:

A. The child rows' foreign key is set to NULL automatically

B. The DBMS prevents deletion of a parent row if matching child rows exist

C. Referential integrity is disabled entirely

D. The parent row is deleted along with all child rows

Ans: B

Explanation

ON DELETE RESTRICT/NO ACTION blocks the deletion of a parent row as long as referencing child rows still exist, thereby preserving referential integrity.

Q104 | **EASY** | **Referential Integrity**

Which of the following column pairs represents a typical primary key-foreign key relationship?

A. STUDENT.RollNo (PK) and ENROLLMENT.RollNo (FK)

B. STUDENT.Address and STUDENT.Phone

C. COURSE.CourseName and COURSE.Duration

D. STUDENT.Name and COURSE.Name

Ans: A

Explanation

STUDENT.RollNo, as the primary key of STUDENT, being referenced as a foreign key in ENROLLMENT, is a standard parent-child key relationship.

Q105 | **DIFFICULT** | **Referential Integrity**

Which of the following is a common misconception about referential integrity?

A. That it applies only to the primary key of a single table

B. That a foreign key can never be NULL under any circumstance

C. That it only concerns insertion, never deletion or update

D. All of the above are common misconceptions

Ans: D

Explanation

All three statements are misconceptions: foreign keys CAN be NULL if allowed; referential integrity concerns the primary-key/foreign-key relationship across two tables, not one table's PK alone; and it applies to inserts, updates, AND deletes.

Q106 | **MODERATE** | **Referential Integrity**

Table ORDERS has CustomerID as a foreign key referencing CUSTOMER(CustomerID). Which of the following updates to ORDERS is INVALID under referential integrity?

A. Setting CustomerID to a value that does not exist in CUSTOMER

B. Setting CustomerID to NULL, if the column allows NULL

C. Setting CustomerID to an existing customer's ID

D. Setting CustomerID to the same value it already had

Ans: A

Explanation

Updating a foreign key to a value with no matching row in the parent table breaks referential integrity.

Q107 | **EASY** | **Referential Integrity**

Which term describes a child-table row whose foreign key does not match any row in the parent table?

A. An orphan row

B. A composite row

C. A normalized row

D. A duplicate row

Ans: A

Explanation

Such a row is called an 'orphan' row because it has lost its valid connection to a parent record.

Q108 | **MODERATE** | **Referential Integrity**

Consider PARENT(ID PK) with rows {1,2,3} and CHILD(ID PK, ParentID FK) with a row having ParentID = 4. What does this represent?

A. An entity integrity violation

B. A valid referential relationship

C. A domain integrity violation

D. A referential integrity violation, since ParentID 4 does not exist in PARENT

Ans: D

Explanation

Since no row with ID = 4 exists in PARENT, the CHILD row's ParentID = 4 is an unmatched foreign key value, violating referential integrity.

Q109 | **DIFFICULT** | **Referential Integrity**

Assertion (A): Referential integrity can be violated only through DELETE operations.
Reason (R): INSERT and UPDATE operations on the foreign key can also introduce unmatched references.
Choose the correct option.

A. A is true and R is false

B. Both A and R are true, and R explains why A is false

C. A is false and R is true

D. Both A and R are false

Ans: B

Explanation

A is false because referential integrity can also be violated by INSERT (adding an unmatched FK) or UPDATE (changing FK to an unmatched value) operations, not only DELETE; R correctly states this, explaining why A is false.

Q110 | **MODERATE** | **Referential Integrity**

Which of the following best distinguishes referential integrity from entity integrity?

A. Referential integrity involves two tables; entity integrity involves a single table's primary key

B. There is no meaningful difference between them

C. Both are only concerned with a single column in a single table

D. Referential integrity is about data types; entity integrity is about triggers

Ans: A

Explanation

Referential integrity concerns the relationship (foreign key to primary key) between two tables, while entity integrity concerns only the primary key within a single table.

Q111 | **EASY** | **Referential Integrity**

In a small EMPLOYEE-DEPARTMENT database, which of these best supports referential integrity?

A. Ensuring every DeptID value in EMPLOYEE exists in DEPARTMENT

B. Allowing duplicate DeptID values in DEPARTMENT

C. Restricting Salary to positive values

D. Making EmpID unique and non-NULL

Ans: A

Explanation

Ensuring that every foreign key value (DeptID in EMPLOYEE) corresponds to an existing primary key value in DEPARTMENT is exactly what referential integrity requires.

Q112 | **MODERATE** | **Referential Integrity**

Which of the following operations on the CHILD table does NOT risk violating referential integrity?

A. Updating a foreign key to an unmatched value

B. Inserting a row with an unmatched foreign key

C. Inserting a row with a foreign key matching an existing parent key

D. Deleting a row from the child table

Ans: D

Explanation

Deleting a row from the child table removes a reference rather than creating an invalid one, so it does not risk a referential integrity violation.

Q113 | **MODERATE** | **Referential Integrity**

A parent row is deleted and the foreign key is defined with ON DELETE SET NULL. What happens to the matching child rows?

A. They are deleted as well

B. The deletion is rejected

C. Their foreign key value becomes 0

D. Their foreign key value becomes NULL

Ans: D

Explanation

SET NULL keeps the child rows but clears the reference, so no orphan reference remains.

Q114 | **DIFFICULT** | **Referential Integrity**

A foreign key uses ON DELETE SET DEFAULT. When a parent row is deleted, matching child rows receive:

A. NULL in all cases

B. the key of the next parent row

C. the column's default value, which must itself match a parent key

D. a copy of the deleted parent row's data

Ans: C

Explanation

SET DEFAULT assigns the column default. To keep referential integrity that default must exist in the parent table (or be NULL).

Q115 | **MODERATE** | **Referential Integrity**

A parent table's primary key value is changed and the foreign key has ON UPDATE CASCADE. What happens to the child rows?

A. Their foreign key is set to NULL

B. Their foreign key values are updated to the new value

C. They are deleted

D. They keep the old value and become orphans

Ans: B

Explanation

ON UPDATE CASCADE carries the key change into all referencing child rows.

Q116 | **MODERATE** | **Referential Integrity**

In EMPLOYEE (EmpID PK, Name, ManagerID FK referring to EmpID), which ManagerID is valid for the head of the organisation, who has no manager?

A. -1

B. NULL

C. The head's own Name

D. 0

Ans: B

Explanation

A self-referencing foreign key may be NULL for a row with no parent. 0 or -1 would need a matching EmpID.

Q117 | **EASY** | **Referential Integrity**

A new department and its first employee are to be inserted (EMPLOYEE refers to DEPARTMENT). Which order preserves referential integrity?

A. Insert the department first, then the employee

B. Either order always works

C. Insert the employee first, then the department

D. Insert both only after deleting existing rows

Ans: A

Explanation

The parent row must exist before a child row can refer to it.

Q118 | **MODERATE** | **Referential Integrity**

With no ON DELETE action defined, how should a department and all its employees be removed without violating referential integrity?

A. Delete both together in any order

B. Delete the employees first, then the department

C. Delete the department first, then the employees

D. Delete only the department; the employees vanish automatically

Ans: B

Explanation

Child rows must go first, otherwise they would briefly refer to a missing parent.

Q119 | **DIFFICULT** | **Referential Integrity**

ENROLL (StudentID, CourseID) is a composite foreign key referring to (StudentID, CourseID) of OFFERING. When is an ENROLL row valid?

A. When the combination matches an existing row of OFFERING

B. When each value appears separately somewhere in OFFERING

C. When the combination is unique within ENROLL

D. When StudentID alone matches an OFFERING row

Ans: A

Explanation

A composite foreign key refers to the combined key, so the whole combination must exist in the parent.

Q120 | **MODERATE** | **Referential Integrity**

Which condition must hold between a foreign key and the primary key it refers to?

A. They must be in the same table

B. They must have compatible data types and the same domain of values

C. They must have identical column names

D. They must both be unique

Ans: B

Explanation

Values are compared for matching, so the two columns need compatible types and domains. Names may differ.

Q121 | **EASY** | **Referential Integrity**

Many employees in EMPLOYEE have DeptID = 10. Does the repeated value violate referential integrity?

A. Yes, but only if DeptID is NULL somewhere

B. No, foreign key values may repeat

C. No, but only if EmpID also repeats

D. Yes, foreign key values must be unique

Ans: B

Explanation

Many child rows may refer to the same parent row; uniqueness is required of the primary key, not of the foreign key.

Q122 | **EASY** | **Referential Integrity**

Must a foreign key column have the same name as the primary key it refers to?

A. No, and the values need not match either

B. Yes, but only for numeric keys

C. Yes, otherwise the DBMS cannot match them

D. No, names may differ; only the values must match

Ans: D

Explanation

The link is made by declaring the reference. Referential integrity depends on matching values, not on names.

Q123 | **EASY** | **Referential Integrity**

Must a foreign key column also be the primary key of its own table?

A. Yes, but only in composite keys

B. No, it must be a second primary key

C. Yes, always

D. No, it is usually an ordinary column of the child table

Ans: D

Explanation

A foreign key only refers to another table's key; it does not have to be the key of its own table.

Q124 | **MODERATE** | **Referential Integrity**

Consider the statements:
I. A foreign key value may be NULL if the column allows it.
II. A foreign key value may repeat in the child table.
III. A non-NULL foreign key value must match a primary key value in the parent table.
IV. A foreign key value must be unique in the child table.
Which statements are correct?

A. II and IV only

B. I, II and III only

C. I, II, III and IV

D. I and III only

Ans: B

Explanation
Foreign key values need not be unique, so IV is wrong. I, II and III describe referential integrity correctly.

Q125 | **MODERATE** | **Referential Integrity**

Assertion (A): In a two-table parent-child relationship, deleting a child row cannot violate referential integrity.
Reason (R): A child row only holds a reference; the parent table does not refer to it.
Choose the correct option.

A. A is false but R is true

B. Both A and R are true, and R explains A

C. Both A and R are true, but R does not explain A

D. A is true but R is false

Ans: B

Explanation
Removing a referencing row leaves no dangling reference, so only deleting a parent row can cause trouble.

Q126 | **MODERATE** | **Referential Integrity**

Which set lists ALL the operations that can violate referential integrity in a parent-child pair of tables?

A. Insert on the parent and delete on the child

B. Only insert on the child

C. Only delete on the parent

D. Insert or update on the child; delete or update on the parent

Ans: D

Explanation
The child can gain an unmatched foreign key by INSERT or UPDATE; the parent can lose or change a referenced key by DELETE or UPDATE.

Q127 | **MODERATE** | **Referential Integrity**

A department currently has no employees. Does this violate referential integrity?

A. Yes, unless the department is deleted

B. No, a parent row may have zero child rows

C. Yes, every parent row needs at least one child

D. No, but only if the department name is NULL

Ans: B

Explanation

Referential integrity checks child rows against parents, not the other way round.

Q128 | **MODERATE** | **Referential Integrity**



FLIGHT (FlightNo, Source, Dest): both Source and Dest are foreign keys referring to AIRPORT (AirportCode), as the diagram shows. Is this allowed?

A. No, Source and Dest must hold the same value

B. Yes, a table may have several foreign keys to the same parent table

C. No, a parent table can be referred to only once

D. Yes, but only if Source and Dest are unique

Ans: B

Explanation

Each foreign key is checked independently against the parent key, so two of them may refer to the same table.

Q129 | **EASY** | **Referential Integrity**



In the diagram, the arrow from ISSUE to BOOK shows that:

A. Every non-NULL BookID in ISSUE must exist in BOOK

B. Every BookID in BOOK must appear in ISSUE

C. ISSUE is the parent table

D. BookID in BOOK is a foreign key of ISSUE

Ans: A

Explanation

The arrow points from the child (holding the foreign key) to the parent (holding the primary key).

Q130 | **EASY** | **Referential Integrity**



Looking at the diagram, which table must already contain a row before a CLASS row can refer to it?

A. Both at the same time

B. TEACHER

C. Neither; order does not matter

D. CLASS

Ans: B

Explanation

TEACHER is the parent table, so its row must exist first.

Q131 MODERATE Referential Integrity

A child row is inserted with a valid foreign key but with a primary key already present in the child table. Which rule is violated?

A. Domain integrity

B. Referential integrity

C. Entity integrity of the child table

D. No rule is violated

Ans: C

Explanation

The foreign key is fine; the duplicated primary key of the child row breaks entity integrity.

Q132 MODERATE Referential Integrity

ON DELETE CASCADE is set on ORDERS.CustomerID. A customer with 3 orders is deleted. How many rows disappear in total?

A. 3: only the orders

B. 0: the deletion is rejected

C. 4: the customer and its 3 orders

D. 1: only the customer

Ans: C

Explanation

CASCADE deletes the referenced parent row and all child rows that refer to it.

Q133 MODERATE Referential Integrity

No ON UPDATE action is defined. What happens if the primary key of a parent row that is still referenced is changed?

A. The update is rejected by the DBMS

B. It succeeds and leaves orphan child rows

C. It succeeds and sets the children's keys to NULL

D. It succeeds and updates the children automatically

Ans: A

Explanation

Without an action such as CASCADE, changing a referenced key would orphan the child rows, so the DBMS refuses it.

SECTION 5

Assertion Concept

Q134 | **EASY** | **Assertion**

An assertion in a database is best described as:

A. A backup mechanism for recovery

B. A row-level constraint on a single column

C. A condition that must always hold true for the database, potentially spanning multiple tables

D. A type of index used to speed up queries

Ans: C

Explanation

An assertion is a general condition or predicate that the database must always satisfy, which can involve one or more tables.

Q135 | **EASY** | **Assertion**

What is the main purpose of using an assertion?

A. To enforce complex, database-wide conditions that simple key constraints cannot express

B. To create backup copies of tables

C. To automatically generate primary keys

D. To encrypt table data

Ans: A

Explanation

Assertions exist to enforce broader, often multi-table conditions that basic entity/domain/referential constraints cannot capture.

Q136 | **MODERATE** | **Assertion**

Which of the following is a suitable use case for an assertion?

A. Restricting a column's data type

B. Ensuring the total number of enrolled students never exceeds the total seats across all courses combined

C. Ensuring a primary key is unique

D. Automatically logging a change to a table

Ans: B

Explanation

A condition spanning multiple tables/rows in aggregate (total enrolled vs total seats) is a classic use case for an assertion, since it cannot be expressed by a simple key or domain constraint.

Q137 | **MODERATE** | **Assertion**

How is an assertion generally different from a simple CHECK constraint on a single column?

A. An assertion can express conditions across multiple tables/rows, while a simple CHECK is usually limited to a single column/row

B. They are functionally identical in all respects

C. An assertion applies only to primary keys

D. A CHECK constraint can span multiple tables but an assertion cannot

Ans: A

Explanation

Assertions are more general and can enforce conditions across multiple tables or aggregated data, unlike a basic single-column CHECK constraint.

Q138 | **EASY** | **Assertion**

At what conceptual level does an assertion typically operate?

A. Single row of a single table

B. Database-wide (may involve multiple tables)

C. Single column of a single table

D. Only on foreign key columns

Ans: B

Explanation

Unlike simple constraints, an assertion is conceptually a database-wide condition, which may reference multiple tables.

Q139 | **MODERATE** | **Assertion**

Which of the following statements about assertions is TRUE?

A. An assertion must always hold true whenever any related data changes

B. Assertions are a synonym for primary keys

C. An assertion is checked only once, when the database is created

D. Assertions can only be used with foreign keys

Ans: A

Explanation

An assertion represents a condition that the DBMS must ensure remains true at all times, being checked whenever relevant data is modified.

Q140 | **EASY** | **Assertion**

Which of these examples best illustrates a possible assertion?

A. "The sum of all department budgets must not exceed the total company budget"

B. "Age must be an integer"

C. "EmpID must be unique"

D. "DeptID must be a foreign key"

Ans: A

Explanation

Ensuring an aggregate condition holds true across the entire database (sum of budgets vs total budget) is a hallmark example of an assertion.

Q141 | **MODERATE** | **Assertion**

Conceptually, an assertion is closest in spirit to which of the following?

A. A user login mechanism

B. A data type definition

C. A general rule/condition that the database must always satisfy, similar to a global invariant

D. A physical storage structure

Ans: C

Explanation

An assertion functions like a global invariant/rule that the whole database must continuously satisfy.

Q142 | **DIFFICULT** | **Assertion**

Which of the following distinguishes an assertion from entity, domain, and referential integrity constraints?

A. Assertions are less powerful and cannot express any condition those constraints already cover

B. Assertions can express more general, often cross-table, business rules beyond the fixed patterns of entity/domain/referential integrity

C. Assertions apply only to primary keys, exactly like entity integrity

D. There is no meaningful difference

Ans: B

Explanation

While entity, domain, and referential integrity address fixed, well-defined situations (keys, values, references), an assertion can express broader, custom business rules, often spanning multiple tables.

Q143 | **MODERATE** | **Assertion**

Which scenario would NOT typically require an assertion (because a simpler constraint already suffices)?

A. Ensuring the number of employees in a department never exceeds a company-wide policy limit involving another table

B. Ensuring one table's aggregate value stays consistent with another table's aggregate value

C. Ensuring a single column's value is always positive

D. Ensuring the combined working hours across two related tables never exceed a fixed total

Ans: C

Explanation

A single-column positivity check can be handled by a simple domain/CHECK constraint; it doesn't need the broader power of an assertion.

Q144 | **EASY** | **Assertion**

Fill in the blank: An assertion is a condition that the database must _____ at all times.

A. delete

B. ignore

C. satisfy (remain true)

D. index

Ans: C

Explanation

By definition, an assertion is a condition that must remain true (be satisfied) at all times within the database.

Q145 | **MODERATE** | **Assertion**

Which of the following best explains why assertions are considered a more powerful integrity mechanism than basic constraints?

A. They are simpler to write than a CHECK constraint

B. They replace the need for a primary key

C. They automatically improve query speed

D. They can express conditions involving multiple tables and aggregated data, not just single-column rules

Ans: D

Explanation

Assertions are more powerful because they can enforce conditions across multiple tables and aggregated values, which basic constraints cannot express.

Q146 | **DIFFICULT** | **Assertion**

Assertion (A): Every business rule in a database can be enforced using only entity, domain, and referential integrity.
Reason (R): Some rules require conditions across multiple tables or aggregated values, which only an assertion (or similar mechanism) can express.
Choose the correct option.

A. Both A and R are false

B. Both A and R are true, and R explains A

C. A is false and R is true

D. A is true and R is false

Ans: C

Explanation

A is false, because some complex, multi-table or aggregate business rules cannot be captured by entity, domain, or referential integrity alone; R correctly explains why such rules need assertions.

Q147 | **MODERATE** | **Assertion**

Which of the following is TRUE regarding the scope of an assertion compared to a table-level constraint?

A. An assertion's scope is always narrower than a single table constraint

B. An assertion cannot reference any table at all

C. An assertion's scope can be broader, potentially spanning the entire database

D. An assertion applies only during database creation, not afterward

Ans: C

Explanation

An assertion is conceptually broader in scope than a typical table-level constraint, as it can span multiple tables across the database.

Q148 | **EASY** | **Assertion**

Which of the following is a key characteristic of an assertion?

A. It only concerns physical storage of data

B. It is only used for formatting output reports

C. It is a synonym for a trigger

D. It expresses a condition that must always evaluate to true for the database to be considered valid

Ans: D

Explanation

An assertion's key characteristic is expressing a condition that must always hold true for the database state to remain valid.

Q149 | **EASY** | **Assertion**

Which two keywords begin the SQL statement used to define an assertion?

A. ADD CONSTRAINT

B. CREATE ASSERTION

C. MAKE CHECK

D. DEFINE RULE

Ans: B

Explanation

An assertion is declared with CREATE ASSERTION, followed by its name and a CHECK condition.

Q150 | **MODERATE** | **Assertion**

At the basic level, an assertion definition consists of:

A. a name and a list of columns

B. a name and a CHECK condition (predicate)

C. a table name and a triggering event

D. a primary key and a foreign key

Ans: B

Explanation

Unlike a trigger, an assertion has no event or action part; it only states a condition that must stay true.

Q151 | **DIFFICULT** | **Assertion**

In a typical assertion, a condition of the form NOT EXISTS (query) expresses that:

A. the query must return every row

B. the tables mentioned must be empty

C. at least one row must exist

D. no row violating the rule may exist in the database

Ans: D

Explanation

The inner query looks for violating rows; NOT EXISTS says there must be none.

Q152 | **MODERATE** | **Assertion**

An INSERT would make the condition of an existing assertion false. What does a DBMS that enforces the assertion do?

A. Rejects the insert

B. Accepts it and silently corrects the data

C. Accepts it and only raises an alert

D. Accepts it and deletes the assertion

Ans: A

Explanation

An assertion must stay true, so any change that would falsify it is refused.

Q153 | **MODERATE** | **Assertion**

Why can assertions be expensive for a DBMS?

A. They must be run manually each night

B. They disable all other constraints

C. They store a duplicate copy of each table

D. The condition may need re-checking whenever any table it mentions changes

Ans: D

Explanation

Because an assertion spans tables and must always hold, relevant updates trigger repeated checking.

Q154 | **DIFFICULT** | **Assertion**

CREATE ASSERTION is part of the SQL standard but many commercial DBMSs do not support it. Which mechanism is commonly used to get a similar effect?

A. Data types alone

B. Primary keys alone

C. Triggers

D. Foreign keys alone

Ans: C

Explanation

Where assertions are not available, rules spanning tables are usually enforced with triggers.

Q155 | **MODERATE** | **Assertion**

Which requirement needs an assertion rather than a simple CHECK on one column?

A. The total salary of a department must not exceed its budget stored in another table

B. Salary must be greater than zero

C. Age must be between 18 and 60

D. Grade must be one of A, B, C or D

Ans: A

Explanation

Only the first rule compares data across rows and tables; the others are single-column domain rules.

Q156 | **DIFFICULT** | **Assertion**

A rule says every department must have at least one employee. Why can a foreign key alone not enforce this?

A. Foreign keys are checked only at table creation

B. Foreign keys cannot refer to primary keys

C. Foreign keys work only on numeric columns

D. A foreign key checks child rows against parents, not whether a parent has children

Ans: D

Explanation

Referential integrity is one-directional. A rule about a parent needing children is a database-wide condition suited to an assertion.

Q157 | **MODERATE** | **Assertion**

Which mechanism states what must be true of the database without specifying an action to take when the condition fails?

A. Default value

B. ON DELETE CASCADE action

C. Trigger

D. Assertion

Ans: D

Explanation

An assertion is a declarative condition; a trigger specifies actions.

Q158 | **MODERATE** | **Assertion**

Assertion (A): An assertion is not tied to one particular table.
Reason (R): Its condition may refer to several tables.
Choose the correct option.

A. A is false but R is true

B. A is true but R is false

C. Both A and R are true, and R explains A

D. Both A and R are true, but R does not explain A

Ans: C

Explanation
Because its condition can span several tables, an assertion belongs to the database rather than to one table.

Q159 | **MODERATE** | **Assertion**

Consider the statements:
I. An assertion can involve more than one table.
II. The DBMS rejects a change that makes an assertion false.
III. An assertion performs a user-defined action when an event occurs.
Which statements are correct?

A. II and III only

B. I and III only

C. I, II and III

D. I and II only

Ans: D

Explanation
III describes a trigger. Assertions only state conditions that must hold.

Q160 | **MODERATE** | **Assertion**

DEPARTMENT

DeptID	Budget
D1	100000

EMPLOYEE

EmpID	DeptID	Salary
1	D1	40000
2	D1	35000

An assertion requires the total salary of a department to be at most its budget. A new employee of D1 with salary 30000 is inserted. What happens?

A. It is accepted, and the oldest employee is deleted

B. It is accepted, as each salary is below the budget

C. It is accepted, and the budget is raised to 105000

D. It is rejected, as the total would be 105000, above 100000

Ans: D

Explanation

40000 + 35000 + 30000 = 105000 exceeds the budget, so the assertion would become false and the insert is refused.

SECTION 6

Trigger Concept

Q161 | **EASY** | **Trigger**

A trigger in a database is best defined as:

A. A set of actions that are automatically executed in response to a specified event on a table

B. A manually executed query written by the user each time

C. A type of primary key

D. A backup file created every night

Ans: A

Explanation

A trigger is a stored program that automatically executes (fires) in response to a specific event, such as an insert, update, or delete on a table.

Q162 | **EASY** | **Trigger**

What is the main purpose of a trigger?

A. To define the primary key of a table

B. To manually validate data once a year

C. To automatically perform an action when a specified data event occurs

D. To physically store backup copies of data

Ans: C

Explanation

Triggers automate actions that should occur whenever a defined event (insert, update, delete) happens on a table.

Q163 | **MODERATE** | **Trigger**

Which of the following events can typically activate a trigger?

A. Only SELECT statements

B. Only when a user logs in for the first time

C. INSERT, UPDATE, or DELETE on a table

D. Only when the database server restarts

Ans: C

Explanation

Triggers are commonly associated with data-modification events: INSERT, UPDATE, and DELETE operations on a table.

Q164 | **EASY** | **Trigger**

A trigger executes its associated actions:

A. Randomly, at unpredictable intervals

B. Once, at database creation time only

C. Only when manually called by the user

D. Automatically, when its specified triggering event occurs

Ans: D

Explanation

The defining feature of a trigger is automatic execution in response to its specified triggering event, without manual invocation.

Q165 | **MODERATE** | **Trigger**

Which basic concept describes how a trigger operates: it fires when an event occurs, checks a condition, and then performs an action?

A. Read-Write-Execute (RWE) concept

B. Insert-Update-Delete (IUD) concept

C. Entity-Domain-Referential (EDR) concept

D. Event-Condition-Action (ECA) concept

Ans: D

Explanation

The Event-Condition-Action (ECA) model describes how a trigger is fired by an event, optionally checks a condition, and then performs an action.

Q166 | **EASY** | **Trigger**

Which of the following is a simple, realistic example of a trigger's use?

A. Manually generating a monthly report by a user

B. Defining a column's data type as INTEGER

C. Automatically updating a 'LastModified' timestamp whenever a row is updated

D. Choosing which column becomes the primary key

Ans: C

Explanation

Automatically updating a timestamp column whenever a row changes is a classic, simple example of a trigger in action.

Q167 | **MODERATE** | **Trigger**

A trigger is defined to fire automatically before a row is deleted from an ORDERS table, logging the deleted row into an ARCHIVE table. Which event activates this trigger?

A. UPDATE on ARCHIVE

B. INSERT on ARCHIVE

C. DELETE on ORDERS

D. SELECT on ORDERS

Ans: C

Explanation

Since the trigger is defined to fire before a row is deleted from ORDERS, the DELETE event on ORDERS is what activates it.

Q168 | **MODERATE** | **Trigger**

Which of the following best describes an automatic action performed by a trigger?

A. The DBMS asks the user for confirmation before every insert

B. The database automatically deletes itself weekly

C. A user manually types a new SQL query every time

D. The system automatically updates a related table when a specified event occurs, without user intervention

Ans: D

Explanation

A trigger's automatic action means the system performs the defined operation (e.g., updating a related table) on its own once the triggering event occurs.

Q169 | **DIFFICULT** | **Trigger**

Which of the following is a common misconception about triggers?

A. That triggers are associated with specific events on a table

B. That triggers can help maintain data consistency automatically

C. That triggers require manual execution by the user every single time, just like a regular query

D. That triggers can fire automatically on INSERT, UPDATE, or DELETE

Ans: C

Explanation

A common misconception is thinking triggers must be manually run like a normal query; in reality, their defining feature is automatic firing on a specified event.

Q170 | **EASY** | **Trigger**

Fill in the blank: A trigger is a set of actions that execute _____ when a specified event occurs on a table.

A. never

B. only during backup

C. automatically

D. manually

Ans: C

Explanation

Automatic execution upon a specified event is the defining characteristic of a trigger.

Q171 | **MODERATE** | **Trigger**

Which of the following scenarios is a good candidate for implementing using a trigger?

A. Automatically maintaining a running total in a SUMMARY table whenever a new SALE record is inserted

B. Restricting a column to only accept integer values

C. Ensuring a primary key is unique

D. Ensuring a foreign key matches an existing primary key

Ans: A

Explanation

Automatically updating a derived/summary value whenever related data changes is a typical, practical use of a trigger, since it involves an automatic reaction to a data event.

Q172 | **DIFFICULT** | **Trigger**

How does a trigger conceptually differ from an assertion?

A. A trigger fires automatically in response to a specific data-modification event, while an assertion is a condition that must always hold true, independent of any specific event

B. An assertion fires only on DELETE events, like a trigger

C. A trigger and an assertion are exactly the same mechanism

D. A trigger can never perform any automatic action

Ans: A

Explanation

A trigger is event-driven and reacts to specific operations (insert/update/delete), while an assertion is a standing condition that the database must always satisfy, not tied to a single triggering event.

Q173 | **MODERATE** | **Trigger**

Which of the following is TRUE about the scope of a basic trigger, as covered at the conceptual (non-advanced) level?

A. It only works with SELECT queries

B. It is generally associated with one specific table and one or more specified events (INSERT/UPDATE/DELETE)

C. It has no relationship with any table at all

D. It can only affect the exact table on which it is defined, and never any other table

Ans: B

Explanation

At the basic conceptual level, a trigger is defined on a specific table for specified events (INSERT/UPDATE/DELETE), even though its actions may sometimes affect other tables.

Q174 | **EASY** | **Trigger**

Which of the following best completes this statement: 'A trigger helps maintain data consistency by...'

A. ...requiring the user to manually check every rule before each transaction

B. ...automatically enforcing actions/rules whenever specified data changes occur

C. ...preventing all changes to the database

D. ...removing the need for any integrity constraints

Ans: B

Explanation

Triggers help maintain consistency by automatically enforcing supplementary actions or rules exactly when relevant data changes happen.

Q175 | **DIFFICULT** | **Trigger**

Assertion (A): A trigger can be used to implement a form of referential-integrity-like behavior, such as cascading a related update.
Reason (R): A trigger automatically executes actions in response to a data-modification event, which can include modifying related tables.
Choose the correct option.

A. Both A and R are true, and R correctly explains A

B. Both A and R are false

C. A is false and R is true

D. A is true and R is false

Ans: A

Explanation

Since a trigger automatically executes actions on related tables in response to events, it can indeed be used to implement cascading-like behavior, and R correctly explains why A is true.

Q176 | **EASY** | **Trigger**

A trigger that executes before the triggering event is applied to the table is called a:

A. FINAL trigger

B. AFTER trigger

C. DEFERRED trigger

D. BEFORE trigger

Ans: D

Explanation

Timing decides whether the trigger runs BEFORE or AFTER the event.

Q177 | **MODERATE** | **Trigger**

A trigger writes a record to an AUDIT table after a new employee row has been inserted. Which timing does it use?

A. BEFORE INSERT

B. BEFORE DELETE

C. AFTER INSERT

D. AFTER DELETE

Ans: C

Explanation

It reacts to an INSERT and runs once the row has been stored, so it is an AFTER INSERT trigger.

Q178 | **MODERATE** | **Trigger**

Which task is best suited to a BEFORE trigger?

A. Recording the old values after an update

B. Counting rows after a delete

C. Checking or adjusting incoming values before the row is stored

D. Logging a row after it was stored

Ans: C

Explanation

A BEFORE trigger sees the data before it is written and can validate or modify it.

Q179 | **DIFFICULT** | **Trigger**

An UPDATE changes 4 rows of a table. How many times do a FOR EACH ROW trigger and a statement-level trigger on UPDATE fire?

A. Row-level: once; statement-level: 4 times

B. Row-level: 4 times; statement-level: once

C. Both: 4 times

D. Both: once

Ans: B

Explanation

A row-level trigger fires for every affected row; a statement-level trigger fires once per statement.

Q180 | **MODERATE** | **Trigger**

In a row-level UPDATE trigger, OLD and NEW refer respectively to:

A. the first and last rows of the table

B. the row values after and before the update

C. the row values before and after the update

D. the parent and child rows

Ans: C

Explanation

OLD holds the values before the change and NEW the values after it.

Q181 | **DIFFICULT** | **Trigger**

Which row references are available in a row-level trigger for each event?

A. All three events provide OLD only

B. INSERT: OLD only; DELETE: NEW only; UPDATE: both

C. INSERT: NEW only; DELETE: OLD only; UPDATE: both OLD and NEW

D. INSERT: both; DELETE: both; UPDATE: NEW only

Ans: C

Explanation

An inserted row has no old version and a deleted row has no new version.

Q182 | **MODERATE** | **Trigger**

Match the ECA parts of a trigger with an example:
(i) Event
(ii) Condition
(iii) Action
(a) Insert a reorder request
(b) Stock falls below 10
(c) A row in ITEM is updated

A. (i)-(c), (ii)-(b), (iii)-(a)

B. (i)-(b), (ii)-(c), (iii)-(a)

C. (i)-(c), (ii)-(a), (iii)-(b)

D. (i)-(a), (ii)-(b), (iii)-(c)

Ans: A

Explanation
The update is the event, 'stock below 10' is the condition tested, and the reorder request is the action.

Q183 | **EASY** | **Trigger**

In the Event-Condition-Action model, what does the condition decide?

A. Which user may define the trigger

B. Whether the table may be updated at all

C. Which event is allowed to occur

D. Whether the action is executed after the event occurs

Ans: D

Explanation
After the event fires the trigger, the condition is tested; only if true does the action run.

Q184 | **MODERATE** | **Trigger**

A trigger fires on every update of Salary, but its condition (new salary more than 1.5 times the old) is false for a given update. What happens?

A. The trigger is deleted

B. The update is cancelled

C. The action runs anyway

D. The event occurs, the condition is checked, and the action is skipped

Ans: D

Explanation
A false condition simply means the action is not executed; the update itself proceeds normally.

Q185 | **EASY** | **Trigger**

A reorder request must be created when Stock falls below 10 after a sale. In the trigger, 'Stock < 10' is the:

A. event

B. condition

C. trigger name

D. action

Ans: B

Explanation

It is the test applied after the event, so it is the condition.

Q186 | **MODERATE** | **Trigger**

A rule says an employee's salary must never be decreased by an update. Which trigger timing is most natural for stopping such a change?

A. AFTER UPDATE

B. BEFORE DELETE

C. BEFORE UPDATE

D. AFTER INSERT

Ans: C

Explanation

A BEFORE trigger can compare OLD and NEW values and stop the change before it is stored.

Q187 | **MODERATE** | **Trigger**

Consider the statements:
I. A trigger can respond to INSERT, UPDATE or DELETE events.
II. A trigger runs only when a user calls it manually.
III. A trigger's action may change rows in another table.
Which statements are correct?

A. II and III only

B. I and II only

C. I and III only

D. I, II and III

Ans: C

Explanation

Triggers run automatically, so II is wrong.

Q188 | **MODERATE** | **Trigger**

Assertion (A): A trigger runs without being called by the user.
Reason (R): The DBMS activates it when the specified event occurs.
Choose the correct option.

A. Both A and R are true, and R explains A

B. A is true but R is false

C. A is false but R is true

D. Both A and R are true, but R does not explain A

Ans: A

Explanation
Automatic, event-driven activation by the DBMS is the defining feature of a trigger.

Q189 | **MODERATE** | **Trigger**

Can a table have more than one trigger?

A. Yes, but only if they use the same event

B. No, only one trigger per table

C. No, unless the table has a foreign key

D. Yes, for different events or timings, such as BEFORE INSERT and AFTER DELETE

Ans: D

Explanation
A table may carry several triggers, each tied to its own event and timing.

Q190 | **EASY** | **Trigger**

Which of the following is a typical use of a trigger?

A. Choosing the primary key of a table

B. Automatically recording changes to an audit table

C. Defining the data type of a column

D. Restricting a column to the values 1 to 12

Ans: B

Explanation
Auditing is a classic trigger task; the others are done by declarations and constraints.

SECTION 7

Comparison Questions

Q191 MODERATE Comparison

Which statement correctly compares entity integrity and domain integrity?

A. Entity integrity governs the primary key's uniqueness/non-NULL status; domain integrity restricts attribute values to a valid set

B. Domain integrity applies only to foreign keys

C. Entity integrity restricts attribute values; domain integrity governs the primary key

D. Both are identical; there is no difference

Ans: A

Explanation

Entity integrity is specifically about the primary key (unique, non-NULL), while domain integrity is about ensuring any attribute's values are valid.

Q192 MODERATE Comparison

Which statement correctly compares entity integrity and referential integrity?

A. Both apply only to foreign keys

B. Entity integrity involves a single table's primary key; referential integrity involves the relationship between a foreign key and a primary key across two tables

C. Entity integrity applies to two tables; referential integrity applies to one

D. There is no meaningful difference between them

Ans: B

Explanation

Entity integrity concerns a single table's primary key, while referential integrity concerns the foreign key-primary key relationship spanning two tables.

Q193 MODERATE Comparison

Which statement correctly compares domain integrity and referential integrity?

A. Both are exactly the same rule

B. Domain integrity concerns two tables; referential integrity concerns one column

C. Referential integrity restricts data types

D. Domain integrity restricts a column's own valid values; referential integrity ensures a foreign key matches a primary key in another table

Ans: D

Explanation

Domain integrity is a single-attribute value restriction, whereas referential integrity governs cross-table foreign key-primary key consistency.

Q194 | **MODERATE** | **Comparison**

Which of the following best compares an assertion and a trigger?

A. An assertion is a condition that must always hold true; a trigger is an action automatically executed in response to a specific event

B. An assertion is event-driven; a trigger is a standing condition

C. Both are identical mechanisms with no difference

D. An assertion can only apply to a single row; a trigger applies to the whole database

Ans: A

Explanation

An assertion represents a standing true-condition for the database, while a trigger is event-driven, firing actions automatically when a specific event occurs.

Q195 | **DIFFICULT** | **Comparison**

Which of the following best compares primary-key-related integrity with foreign-key-related integrity?

A. Foreign-key-related integrity concerns uniqueness within its own table only

B. Primary-key-related integrity concerns matching values in another table

C. Both concern only a single table and ignore relationships

D. Primary-key-related integrity (entity integrity) ensures uniqueness/non-NULL of the key within its own table; foreign-key-related integrity (referential integrity) ensures the key correctly references another table's primary key

Ans: D

Explanation

Primary-key-related integrity (entity integrity) is self-contained within one table, while foreign-key-related integrity (referential integrity) inherently involves a relationship with another table.

Q196 | **MODERATE** | **Comparison**

Which of the following correctly compares a basic constraint (entity/domain/referential) with an assertion?

A. An assertion can never involve more than one table

B. A basic constraint is always broader than an assertion

C. There is no functional difference

D. A basic constraint typically applies to a fixed, well-defined pattern; an assertion can express broader, custom conditions, often across multiple tables

Ans: D

Explanation

Basic constraints follow fixed, well-known patterns (key uniqueness, valid values, valid references), while assertions offer more general, custom, often cross-table condition enforcement.

Q197 | **EASY** | **Comparison**

Which pair below correctly matches an integrity rule to the key it primarily concerns?

A. Referential integrity - Primary key of the same table only

B. Entity integrity - Primary key

C. Domain integrity - Foreign key

D. Entity integrity - Foreign key

Ans: B

Explanation

Entity integrity is specifically defined around the primary key of a table.

Q198 | **DIFFICULT** | **Comparison**

Which of the following statements comparing trigger and assertion is MOST accurate?

A. An assertion always fires on INSERT events like a trigger

B. A trigger reacts to an event and can perform actions; an assertion passively defines a condition that the DBMS checks for continued validity, without itself performing corrective actions

C. Triggers and assertions serve identical purposes with no distinction

D. A trigger cannot perform any action, only assertions can

Ans: B

Explanation

A trigger is proactive and event-driven, performing defined actions, whereas an assertion is more passive — it defines a condition the DBMS must keep true, without itself specifying corrective actions.

Q199 | **MODERATE** | **Comparison**

Match the integrity type with its primary concern:
(i) Entity integrity
(ii) Domain integrity
(iii) Referential integrity
(a) Cross-table key consistency
(b) Primary key uniqueness/non-NULL
(c) Valid attribute values

A. (i)-(b), (ii)-(a), (iii)-(c)

B. (i)-(a), (ii)-(b), (iii)-(c)

C. (i)-(b), (ii)-(c), (iii)-(a)

D. (i)-(c), (ii)-(a), (iii)-(b)

Ans: C

Explanation

Entity integrity matches primary key uniqueness/non-NULL (b), domain integrity matches valid attribute values (c), and referential integrity matches cross-table key consistency (a).

Q200 | **MODERATE** | **Comparison**

Which of the following is true when comparing all five topics: entity, domain, referential integrity, assertion, and trigger?

A. Assertions and triggers are unrelated to integrity at all

B. Only triggers can be automatically enforced; the rest require manual checking

C. Entity, domain, and referential integrity are declarative rules enforced automatically by key/value constraints; assertions declare broader conditions; triggers are active, event-driven procedures

D. All five are exactly the same mechanism

Ans: C

Explanation

Entity, domain, and referential integrity are foundational declarative constraints; assertions extend this to broader declarative conditions; triggers add active, event-driven, automatic behavior — together forming the full integrity toolkit covered in this syllabus.

Q201 | **MODERATE** | **Comparison**

Which statement correctly compares a primary key and a foreign key?

A. Foreign key values must be unique; primary key values may repeat

B. Both must be unique and non-NULL

C. Both may repeat but neither may be NULL

D. Primary key values are unique in their table; foreign key values may repeat

Ans: D

Explanation

Entity integrity demands unique, non-NULL primary keys. Foreign key values may repeat and may be NULL if allowed.

Q202 | **DIFFICULT** | **Comparison**

EMPLOYEE.DeptID is an INTEGER foreign key referring to DEPARTMENT. Which value violates domain integrity rather than referential integrity?

A. 999, which is not in DEPARTMENT

B. 10, which exists in DEPARTMENT

C. 'ABC'

D. NULL, when the column allows NULL

Ans: C

Explanation

'ABC' is not an integer, a data-type (domain) violation. 999 would break referential integrity; 10 and NULL are acceptable.

Q203 | **MODERATE** | **Comparison**

Match each requirement with the mechanism that fits best:
(i) Marks between 0 and 100
(ii) OrderID must exist in ORDERS
(iii) Stock updated automatically after a sale
(iv) Total loans of a branch limited by its deposits
(a) Trigger
(b) Referential integrity
(c) Assertion
(d) Domain integrity

A. (i)-(a), (ii)-(b), (iii)-(d), (iv)-(c)

B. (i)-(d), (ii)-(b), (iii)-(c), (iv)-(a)

C. (i)-(d), (ii)-(b), (iii)-(a), (iv)-(c)

D. (i)-(b), (ii)-(d), (iii)-(a), (iv)-(c)

Ans: C

Explanation
A range rule is domain integrity, a reference to another table is referential integrity, an automatic update is a trigger and a multi-table limit is an assertion.

Q204 | **MODERATE** | **Comparison**

What is the main difference between a constraint such as a foreign key and a trigger?

A. A constraint only allows or rejects a change; a trigger can perform further actions

B. There is no difference in what they can do

C. A constraint fires on events; a trigger never does

D. A trigger only checks key values; a constraint does not

Ans: A

Explanation
Constraints accept or reject. Triggers can also carry out actions such as logging or updating other tables.

Q205 | **EASY** | **Comparison**

A NULL in a primary key and the value -5 in an Age column (valid range 18 to 60) violate, respectively:

A. entity integrity and domain integrity

B. entity integrity and referential integrity

C. referential integrity and domain integrity

D. domain integrity and entity integrity

Ans: A

Explanation
NULL in the primary key breaks entity integrity; -5 is outside the allowed range, a domain violation.

Q206 | **EASY** | **Comparison**

Which integrity rule must examine TWO tables to decide whether a change is valid?

A. Domain integrity

B. Referential integrity

C. Entity integrity on a composite key

D. Entity integrity

Ans: B

Explanation

Referential integrity compares a foreign key in one table with the primary key of another.

Q207 | **MODERATE** | **Comparison**

Which statement about assertions and referential integrity is correct?

A. Referential integrity applies to one column; an assertion to one table

B. A foreign key can state every rule an assertion can

C. An assertion is a special kind of primary key

D. An assertion can state rules a foreign key cannot, such as every parent having a child

Ans: D

Explanation

Foreign keys only link child rows to parents. Broader database-wide conditions need assertions.

SECTION 8

Table-Based Integrity Questions

Q208 | **MODERATE** | **Table-Based**

STUDENT table:

RollNo	Name	Age
101	Aman	17
102	Riya	18
101	Suresh	19

Which integrity rule is violated?

A. Referential integrity, because of a missing foreign key

B. Entity integrity, because RollNo 101 is duplicated

C. Domain integrity, because Age is invalid

D. No rule is violated

Ans: B

Explanation

RollNo, the primary key, repeats the value 101 in two rows, violating the uniqueness requirement of entity integrity.

Q209 | **MODERATE** | **Table-Based**

EMPLOYEE table:

EmpID	Name	Age
201	Kavita	25
NULL	Manoj	30
203	Sneha	28

Which integrity rule is violated?

A. Referential integrity

B. No violation, since NULL is always allowed

C. Entity integrity, since a primary key value is NULL

D. Domain integrity

Ans: C

Explanation

EmpID is the primary key, and a NULL value in a primary key column directly violates entity integrity.

Q210 MODERATE Table-Based

STUDENT table:

RollNo	Name	Age
1	Priya	16
2	Rohit	-5
3	Sonal	20

Which integrity rule is violated, assuming Age must be a positive number?

A. Domain integrity, since Age = -5 is invalid

B. Referential integrity

C. Entity integrity

D. No violation

Ans: A

Explanation

A negative age value falls outside the valid domain for Age, which must be positive, so this is a domain integrity violation.

Q211 MODERATE Table-Based

DEPARTMENT (parent):

DeptID	DeptName
1	CS
2	Maths

EMPLOYEE (child):

EmpID	Name	DeptID
101	Aman	1
102	Riya	3

Which integrity rule is violated?

A. Referential integrity, since DeptID = 3 does not exist in DEPARTMENT

B. Entity integrity in EMPLOYEE

C. Domain integrity in DEPARTMENT

D. No violation

Ans: A

Explanation

EMPLOYEE's foreign key DeptID = 3 has no matching row in the parent DEPARTMENT table, violating referential integrity.

Q212 | **EASY** | **Table-Based**

BOOK table:

BookID	Title	Price
B01	DBMS Basics	250
B02	Data Structures	300
B03	Networks	275

Does this table satisfy entity integrity?

A. No, because BookID values repeat

B. Yes, BookID values are unique and non-NULL

C. No, because Price is negative

D. Cannot be determined

Ans: B

Explanation

All BookID values are unique and none are NULL, so the table satisfies entity integrity.

Q213 | **MODERATE** | **Table-Based**

COURSE table:

CourseID	CourseName	Seats
C1	DBMS	60
C2	OS	'Sixty'
C3	CN	50

Assuming Seats must be an integer, which value violates domain integrity?

A. None of the values violate domain integrity

B. 60 in row C1

C. 'Sixty' in row C2

D. 50 in row C3

Ans: C

Explanation

The text value 'Sixty' does not match the required integer data type for Seats, violating domain integrity.

Q214 | **MODERATE** | **Table-Based**

PARENT table: ID(PK) = {10, 20, 30} CHILD table: ID(PK), ParentID(FK) with a row ParentID = 20.

Is this CHILD row's foreign key value valid?

A. No, foreign keys are never valid

B. Cannot be determined without more information

C. Yes, because 20 exists as an ID in PARENT

D. No, because 20 must also exist in CHILD

Ans: C

Explanation

Since 20 exists as a valid primary key value in PARENT, the CHILD row's foreign key reference is valid under referential integrity.

Q215 | **MODERATE** | **Table-Based**

ORDER table (child) references CUSTOMER (parent) via CustomerID. A new ORDER row is inserted with CustomerID = NULL, and the column allows NULL. Is this insertion valid?

A. Yes, but only if CustomerID is the primary key of ORDER

B. No, because NULL always violates entity integrity

C. No, NULL is never allowed under referential integrity

D. Yes, since a NULL foreign key (if permitted by the column) does not violate referential integrity

Ans: D

Explanation

Referential integrity allows a foreign key to be NULL when the column permits it, representing an optional (unassigned) relationship.

Q216 DIFFICULT Table-Based

STUDENT table:

RollNo	Name	Email
1	Meena	meena@mail.com
2	Arjun	arjun.mail.com
3	Divya	divya@mail.com

Assuming Email must contain '@', which insertion violates the defined domain?

A. Row 3, Divya's email

B. Row 2, Arjun's email, since it lacks '@'

C. Row 1, Meena's email

D. None of the rows violate the domain

Ans: B

Explanation

Arjun's email 'arjun.mail.com' does not contain the required '@' symbol, violating the defined format restriction under domain integrity.

Q217 MODERATE Table-Based

DEPARTMENT (parent) has DeptID = {1,2,3}. Which DELETE operation on DEPARTMENT would create a referential integrity issue if EMPLOYEE (child) still has rows referencing DeptID = 2, and no cascade rule is set?

A. Deleting a row that does not exist

B. Inserting a new department

C. Deleting the row where DeptID = 2, while employees still reference it

D. Deleting the row where DeptID = 1, if no employee references it

Ans: C

Explanation

Deleting DeptID = 2 while EMPLOYEE rows still reference it (with no ON DELETE rule) would leave those employees pointing to a non-existent department, violating referential integrity.

Q218 | **MODERATE** | **Table-Based**

STUDENT table:

RollNo(PK)	Name	Age
1	Kunal	15
2	Priti	16
3	Kunal	15

Does this table violate entity integrity?

A. Yes, because names repeat

B. No, because RollNo values (1,2,3) are all unique, even though names repeat

C. Yes, because RollNo repeats

D. Cannot be determined

Ans: B

Explanation

Entity integrity only concerns the primary key (RollNo here), which is unique and non-NULL in every row; repeated names in a non-key column do not violate entity integrity.

Q219 | **DIFFICULT** | **Table-Based**

TEACHER (parent): TeacherID = {T1, T2} CLASS (child): ClassID(PK), TeacherID(FK)

ClassID	TeacherID
C101	T1
C102	T2
C103	T1

A new row is inserted: C104	T3. Which rule is violated, and why?

A. Entity integrity, because ClassID repeats

B. Referential integrity, because T3 does not exist in TEACHER

C. Domain integrity, because ClassID is invalid

D. No violation occurs

Ans: B

Explanation

TeacherID = T3 has no matching row in the parent TEACHER table, so this insertion violates referential integrity.

Q220 | **MODERATE** | **Table-Based**

STOCK table:

ItemCode(PK)	Quantity
I1	50
I2	-10
I3	20

Assuming Quantity cannot be negative, which type of violation occurs in row I2, and which constraint should catch it?

A. Domain integrity violation, caught by a CHECK constraint (Quantity >= 0)

B. Entity integrity violation, caught by a UNIQUE constraint

C. No violation occurs

D. Referential integrity violation, caught by a FOREIGN KEY constraint

Ans: A

Explanation

A negative Quantity breaks the valid-value domain for that attribute; this is a domain integrity violation, typically caught using a CHECK constraint.

Q221 | **DIFFICULT** | **Table-Based**

SUPPLIER (parent): SupplierID = {S1, S2, S3} PRODUCT (child): ProductID(PK), SupplierID(FK)

ProductID	SupplierID
P1	S1
P2	S2

Which of the following updates to PRODUCT would violate referential integrity?

A. Changing P2's ProductID to P3

B. Changing P1's SupplierID from S1 to S3

C. Inserting P3 with SupplierID = S2

D. Changing P1's SupplierID from S1 to S9, which does not exist in SUPPLIER

Ans: D

Explanation

Changing the foreign key to S9, a value absent from the parent SUPPLIER table, creates an unmatched reference and violates referential integrity.

Q222 | **MODERATE** | **Table-Based**

Identify the correct integrity constraint that governs the following requirement: 'The CustomerID in the ORDER table must always correspond to an existing customer in the CUSTOMER table.'

A. Referential integrity

B. Domain integrity

C. Entity integrity

D. Assertion only

Ans: A

Explanation

This is a textbook description of referential integrity: a foreign key (CustomerID) must correspond to a valid primary key in the referenced table.

Q223 | **MODERATE** | **Table-Based**

ENROLL (composite primary key: StudentID, CourseID)

StudentID	CourseID
1	C1
1	C2
2	C1

Does the table satisfy entity integrity?

A. No, because CourseID C1 appears twice

B. It cannot be decided without a foreign key

C. No, because StudentID 1 appears twice

D. Yes, every (StudentID, CourseID) combination is unique and non-NULL

Ans: D

Explanation

In a composite key only the combination must be unique. Repeated parts are allowed.

Q224 | **MODERATE** | **Table-Based**

ENROLL (composite primary key: StudentID, CourseID)

StudentID	CourseID
1	C1
1	C2
1	C1

Which rule is violated?

A. Referential integrity, as StudentID repeats

B. Domain integrity, as CourseID repeats

C. No rule is violated

D. Entity integrity, as the combination (1, C1) is repeated

Ans: D

Explanation

The first and third rows have the same composite key, breaking uniqueness.

Q225 | **MODERATE** | **Table-Based**

ENROLL (composite primary key: StudentID, CourseID)

StudentID	CourseID
1	C1
2	NULL

Which statement is correct?

A. Row 2 violates entity integrity because part of the key is NULL

B. Both rows are valid

C. Row 2 violates referential integrity

D. Row 1 violates entity integrity

Ans: A

Explanation

No attribute of the primary key may be NULL.

Q226 MODERATE Table-Based

STUDENT (primary key: RollNo)

RollNo	Name
1	Asha
2	Ravi
2	Meena
NULL	Kiran

Which violations are present?

A. No violation, since the names differ

B. Both a duplicate RollNo and a NULL RollNo

C. Only a NULL RollNo

D. Only a duplicate RollNo

Ans: B

Explanation

RollNo 2 is repeated and one RollNo is NULL; both break entity integrity.

Q227 MODERATE Table-Based

PRODUCT (rule: Price must be greater than 0)

Code	Price
P1	50
P2	0
P3	-3
P4	12

How many rows violate domain integrity?

A. One (P3 only)

B. None

C. Three

D. Two (P2 and P3)

Ans: D

Explanation

0 and -3 are not greater than 0, so P2 and P3 are outside the domain.

Q228 | **MODERATE** | **Table-Based**

BIRTH (rule: Month must be an integer from 1 to 12)

ID	Month
1	3
2	12
3	13
4	0

Which IDs violate domain integrity?

A. 1 and 2

B. Only 3

C. None of them

D. 3 and 4

Ans: D

Explanation

13 and 0 lie outside the range 1 to 12.

Q229 | **EASY** | **Table-Based**

PERSON (rule: Gender must be M or F)

ID	Gender
1	M
2	F
3	Male
4	F

Which row violates domain integrity?

A. The row with ID 2

B. The row with ID 3

C. The row with ID 1

D. The row with ID 4

Ans: B

Explanation

'Male' is not one of the allowed values M or F.

Q230 | **MODERATE** | **Table-Based**

EMP (primary key: EmpID; rule: DOB must be a valid date)

EmpID	DOB
1	2001-04-10
2	2003-13-05
3	1999-07-21

Which row violates domain integrity?

A. None, since all follow the same format

B. EmpID 2, as month 13 does not exist

C. EmpID 1

D. EmpID 3

Ans: B

Explanation

2003-13-05 is not a valid date because there is no month 13.

Q231 | **MODERATE** | **Table-Based**

EMPLOYEE (primary key: EmpID; ManagerID refers to EmpID)

EmpID	Name	ManagerID
1	Asha	NULL
2	Ravi	1
3	Meena	7

Which row violates referential integrity?

A. EmpID 3, since manager 7 does not exist

B. EmpID 1, since ManagerID is NULL

C. None of the rows

D. EmpID 2, since 1 is also an EmpID

Ans: A

Explanation

ManagerID 7 matches no EmpID. NULL is allowed and 1 exists.

Q232 | **DIFFICULT** | **Table-Based**

DEPT (DeptID): 1, 2, 3

EMP (DeptID refers to DEPT; NULL allowed)

EmpID	DeptID
10	1
11	NULL
12	2
13	5

How many EMP rows violate referential integrity?

A. Two: EmpIDs 11 and 13

B. Three

C. None

D. Exactly one: EmpID 13

Ans: D

Explanation

DeptID 5 has no parent. The NULL in row 11 is allowed.

Q233 | **EASY** | **Table-Based**

DEPARTMENT (DeptID): 10, 20, 30

EMPLOYEE (DeptID refers to DEPARTMENT; deletions of referenced rows are restricted)

EmpID	DeptID
1	10
2	10
3	20

Which department can be deleted without any problem?

A. DeptID 30

B. DeptID 10

C. DeptID 20

D. Any department

Ans: A

Explanation

Only DeptID 30 is not referenced by any employee.

Q234 | **MODERATE** | **Table-Based**

DEPARTMENT (DeptID): 10, 20

EMPLOYEE (DeptID refers to DEPARTMENT, ON UPDATE CASCADE)

EmpID	DeptID
1	10
2	20
3	10

DeptID 10 in DEPARTMENT is changed to 100. What do the EMPLOYEE rows show afterwards?

A. EmpIDs 1 and 3 become NULL

B. EmpIDs 1 and 3 are deleted

C. EmpIDs 1 and 3 have DeptID 100; EmpID 2 keeps 20

D. All three rows keep their old DeptID

Ans: C

Explanation

CASCADE copies the new key to all rows that referred to the old one.

Q235 | **MODERATE** | **Table-Based**

CUSTOMER (CustID): C1, C2

ORDERS (CustID refers to CUSTOMER, ON DELETE SET NULL)

OrderID	CustID
O1	C1
O2	C1
O3	C2

Customer C1 is deleted. What do the ORDERS rows show?

A. O1 and O2 have CustID NULL; O3 is unchanged

B. O1 and O2 are deleted

C. The deletion is rejected

D. O1, O2 and O3 all have CustID NULL

Ans: A

Explanation

SET NULL clears only the foreign keys that referred to the deleted customer.

Q236 | **MODERATE** | **Table-Based**

CUSTOMER (CustID): C1, C2

ORDERS (CustID refers to CUSTOMER, ON DELETE CASCADE)

OrderID	CustID
O1	C1
O2	C1
O3	C2

Customer C1 is deleted. How many rows remain in ORDERS?

A. 0

B. 1

C. 2

D. 3

Ans: B

Explanation

CASCADE removes O1 and O2 with the customer, so only O3 remains.

Q237 | **MODERATE** | **Table-Based**

ORDERS (primary key: OrderID; CustID refers to CUSTOMER)

CUSTOMER (CustID): C1, C2, C3

OrderID	CustID
O1	C1
O2	C2

Which new row can be inserted without violating any integrity rule?

A. O3 | C9

B. O1 | C1

C. O3 | C2

D. NULL | C3

Ans: C

Explanation

O3 | C9 breaks referential integrity; O1 | C1 and a NULL OrderID both break entity integrity.

Q238 | **DIFFICULT** | **Table-Based**

DEPT (DeptID): D1, D2

STUDENT (primary key: RollNo; Age must be 18 to 25; DeptID refers to DEPT)

RollNo	Age	DeptID
1	20	D1
2	40	D1
3	22	D9

Which statement is correct?

A. Only row 3 violates an integrity rule

B. Row 2 violates referential integrity and row 3 violates domain integrity

C. Rows 2 and 3 both violate entity integrity

D. Row 2 violates domain integrity and row 3 violates referential integrity

Ans: D

Explanation

Age 40 is outside 18 to 25 (domain); D9 has no parent in DEPT (referential).

Q239 | **MODERATE** | **Table-Based**

DEPARTMENT (DeptID): 1, 2, 3

EMPLOYEE (DeptID refers to DEPARTMENT; no ON UPDATE action)

EmpID	DeptID
10	1
11	2

Which update is invalid?

A. Changing DeptID 1 in DEPARTMENT to 10

B. Changing the DeptID of EmpID 11 to 1

C. Changing the name of department 2

D. Changing DeptID 3 in DEPARTMENT to 30

Ans: A

Explanation

DeptID 1 is still referenced by EmpID 10. DeptID 3 is unreferenced and the other changes keep valid references.

Q240 | **EASY** | **Table-Based**

STUDENT (primary key: RollNo; Marks must be 0 to 100)

RollNo	Marks
101	78
102	64
103	91

A row (104, 150) is inserted. Which rule stops it?

A. Assertion

B. Entity integrity

C. Referential integrity

D. Domain integrity

Ans: D

Explanation

RollNo 104 is new, so entity integrity holds. Marks 150 is outside 0 to 100.

Q241 | **DIFFICULT** | **Table-Based**

STUDENT (StudentID): S1, S2, S3
COURSE (CourseID): C1, C2

ENROLL (StudentID and CourseID are foreign keys)

StudentID	CourseID
S1	C1
S2	C9
S5	C2

Which ENROLL rows violate referential integrity?

A. None of the rows

B. Only the third row

C. The second and third rows

D. Only the second row

Ans: C

Explanation

C9 is not in COURSE and S5 is not in STUDENT.

Q242 | **MODERATE** | **Table-Based**

BOOK (BookID): B1, B2, B3

ISSUE (BookID refers to BOOK)

IssueID	BookID
1	B1
2	B3

Book B2 is deleted. Is referential integrity violated?

A. Yes, because B2 is a parent row

B. Yes, because ISSUE has fewer rows than BOOK

C. No, but only if B1 and B3 are also deleted

D. No, because no ISSUE row refers to B2

Ans: D

Explanation

Deleting a parent row is a problem only when child rows still refer to it.

SECTION 9

Scenario-Based Questions

Q243 MODERATE Scenario

A school database assigns each student a unique AdmissionNo as the primary key. A new admission clerk accidentally leaves the AdmissionNo field blank while entering a new student. Which rule prevents this record from being saved?

A. Assertion

B. Domain integrity

C. Entity integrity

D. Referential integrity

Ans: C

Explanation

Since AdmissionNo is the primary key, entity integrity requires it to be non-NULL, so the DBMS should reject the blank entry.

Q244 MODERATE Scenario

A hospital database restricts the 'BloodGroup' column to only 8 valid values (A+, A-, B+, B-, O+, O-, AB+, AB-). A nurse tries to enter 'AB++' by mistake. Which rule blocks this entry?

A. Trigger

B. Domain integrity

C. Entity integrity

D. Referential integrity

Ans: B

Explanation

'AB++' is not part of the valid set of BloodGroup values, so domain integrity prevents this invalid entry.

Q245 MODERATE Scenario

An online store has ORDER records referencing CustomerID from the CUSTOMER table. A customer account is deleted while they still have pending orders, and no cascading rule exists. What integrity problem results?

A. No problem occurs

B. Entity integrity violation in CUSTOMER

C. Domain integrity violation in ORDER

D. Referential integrity violation, since ORDER rows now reference a non-existent customer

Ans: D

Explanation

Deleting the referenced parent row while child ORDER rows still reference it (without cascading) leaves those orders pointing to a non-existent customer, violating referential integrity.

Q246 | **MODERATE** | **Scenario**

A bank wants to ensure that whenever a withdrawal is made, the account balance is automatically checked and an alert is generated if it falls below the minimum balance. Which mechanism is most suitable?

A. Referential integrity

B. Entity integrity

C. A trigger

D. Domain integrity

Ans: C

Explanation

Automatically performing an action (an alert) in response to a data-modification event (withdrawal) is a classic use case for a trigger.

Q247 | **MODERATE** | **Scenario**

A university wants to enforce a rule that the total number of students enrolled across all sections of a course must never exceed the course's total capacity, where enrollment data is spread across multiple tables. Which mechanism is most appropriate?

A. An assertion

B. A simple CHECK constraint on one column

C. Entity integrity

D. Domain integrity

Ans: A

Explanation

Since this rule spans multiple tables/aggregated data rather than a single column, an assertion is the most conceptually appropriate mechanism.

Q248 | **EASY** | **Scenario**

A library assigns every book a unique BookID, and no two books can share the same ID. Which integrity rule ensures this?

A. Entity integrity

B. Trigger

C. Referential integrity

D. Domain integrity

Ans: A

Explanation

Ensuring the primary key (BookID) is unique for every book is precisely what entity integrity guarantees.

Q249 MODERATE Scenario

An e-commerce system automatically updates a product's 'Stock' count whenever a new sale is recorded, without any manual action from staff. Which mechanism enables this automatic behavior?

A. Entity integrity

B. Domain integrity

C. A trigger

D. Referential integrity

Ans: C

Explanation

Automatically updating Stock in response to the event of a new sale being recorded is a typical trigger-based behavior.

Q250 MODERATE Scenario

A payroll system stores EmployeeID as a foreign key in the SALARY table, referencing EMPLOYEE. A payroll clerk tries to insert a salary record for EmployeeID = 9999, which does not exist in EMPLOYEE. What happens if referential integrity is enforced?

A. Referential integrity does not apply to SALARY records

B. The insertion is rejected, since EmployeeID 9999 does not exist in the parent table

C. The insertion succeeds silently with a NULL value substituted

D. The insertion succeeds and a new employee is auto-created

Ans: B

Explanation

A DBMS enforcing referential integrity rejects any child insertion whose foreign key value does not exist in the parent table.

Q251 MODERATE Scenario

A college wants Age of students to always be between 16 and 25. Which mechanism best enforces this at the column level?

A. An assertion is strictly required, no simpler mechanism will work

B. Domain integrity, using a range restriction

C. Referential integrity

D. Entity integrity

Ans: B

Explanation

Restricting a single column's values to a valid numeric range is a straightforward application of domain integrity, typically via a CHECK constraint.

Q252 | **DIFFICULT** | **Scenario**

A company wants to automatically log every deletion from the EMPLOYEE table into an AUDIT table, capturing who deleted the record and when, without requiring the user to do this manually. Which mechanism should be used?

A. A trigger fired on the DELETE event

B. Entity integrity

C. Referential integrity

D. Domain integrity

Ans: A

Explanation

Automatically logging details whenever a DELETE event occurs is a textbook use case for a trigger fired on that event.

Q253 | **MODERATE** | **Scenario**

A hospital's PATIENT table uses PatientID as primary key, and the APPOINTMENT table references PatientID as a foreign key. If an appointment is scheduled for a PatientID that has never been registered, which rule is violated?

A. Entity integrity in APPOINTMENT

B. Referential integrity, since the PatientID does not exist in PATIENT

C. No rule is violated

D. Domain integrity in PATIENT

Ans: B

Explanation

Scheduling an appointment with a foreign key value (PatientID) that has no corresponding row in the parent PATIENT table violates referential integrity.

Q254 | **DIFFICULT** | **Scenario**

A university enforces a rule that no student can be enrolled in more than 6 courses in a single semester, where enrollment counts must be checked across the entire ENROLLMENT table for that student. Which mechanism most naturally expresses this database-wide condition?

A. Referential integrity alone

B. An assertion checking the aggregated enrollment count

C. Entity integrity

D. Domain integrity on a single column

Ans: B

Explanation

Since the rule requires evaluating an aggregated condition (count of enrollments) that spans multiple rows, it is best expressed as an assertion.

Q255 | **MODERATE** | **Scenario**

A hospital deletes a doctor's record but wants past appointments to remain, showing no doctor assigned. Which referential action suits this?

A. ON DELETE CASCADE

B. No action; the appointments adjust themselves

C. ON DELETE RESTRICT

D. ON DELETE SET NULL

Ans: D

Explanation

SET NULL keeps the appointment rows and clears their reference to the deleted doctor.

Q256 | **MODERATE** | **Scenario**

A college removes a department and wants all its courses removed automatically. Which referential action should the foreign key have?

A. ON DELETE CASCADE

B. ON DELETE RESTRICT

C. ON UPDATE CASCADE

D. ON DELETE SET NULL

Ans: A

Explanation

CASCADE deletes the child rows together with the parent row.

Q257 | **EASY** | **Scenario**

During a data migration, 20 legacy student rows have no ID, and ID is the table's primary key. What does a DBMS enforcing entity integrity do?

A. Inserts them with NULL IDs

B. Inserts them with ID 0 automatically

C. Inserts them but stops checking duplicates

D. Rejects those rows

Ans: D

Explanation

A primary key cannot be NULL, so rows without an ID are refused.

Q258 MODERATE Scenario

A store requires Stock to be zero or more at all times for each item. Which mechanism most directly expresses this single-column rule?

A. An assertion across all tables

B. Referential integrity

C. Entity integrity

D. Domain integrity (a range rule)

Ans: D

Explanation

A rule on one attribute's values is a domain restriction.

Q259 MODERATE Scenario

Whenever a student's marks are updated, the old marks must be saved automatically in a MARKS_HISTORY table. Which mechanism does this need?

A. An AFTER UPDATE trigger using the OLD values

B. A domain restriction on Marks

C. A foreign key from MARKS_HISTORY

D. An assertion on the Marks column

Ans: A

Explanation

It is an automatic action on an update event that needs the previous values, which is what a trigger provides.

Q260 MODERATE Scenario

A bank requires each branch's total loan amount to stay below ten times its total deposits, which are kept in other tables. Which mechanism fits best?

A. A default value

B. A domain restriction on Amount

C. Entity integrity on BranchID

D. An assertion

Ans: D

Explanation

The rule compares aggregates across tables, a database-wide condition.

Q261 | **MODERATE** | **Scenario**

A school database must satisfy: (a) unique roll numbers, (b) marks 0 to 100, (c) every enrolment refers to an existing student, (d) an automatic email on admission. Which mapping is correct?

A. (a) domain, (b) entity, (c) referential, (d) trigger

B. (a) entity, (b) domain, (c) trigger, (d) referential

C. (a) entity, (b) domain, (c) referential, (d) trigger

D. (a) entity, (b) referential, (c) domain, (d) assertion

Ans: C

Explanation

Unique keys are entity integrity, a value range is domain integrity, a reference is referential integrity and an automatic action is a trigger.

Q262 | **MODERATE** | **Scenario**

Student names must be converted to capital letters automatically before each new student row is saved. Which mechanism suits this?

A. An assertion

B. Entity integrity

C. Domain integrity

D. A BEFORE INSERT trigger

Ans: D

Explanation

Changing data automatically before it is stored is a BEFORE trigger task.

Q263 | **EASY** | **Scenario**

A scanner records ISBN codes in the LOAN table, and each code must already exist in the BOOK table. Which integrity rule enforces this?

A. Domain integrity

B. Referential integrity

C. Entity integrity

D. An assertion only

Ans: B

Explanation

A value in one table must match a key in another table.

Q264	DIFFICULT	Scenario

A hospital must reject a prescription if the patient's allergy list, kept in another table, contains the prescribed drug. Which mechanism can check this on every new prescription?

A. Referential integrity on PatientID

B. A BEFORE INSERT trigger on PRESCRIPTION

C. Entity integrity on PrescriptionID

D. A domain restriction on the Drug column

Ans: B

Explanation

The check needs data from another table and must run on each insert, which a trigger can do. The other options cannot compare against the allergy table.

SECTION 10

Difficult & Tricky Questions

Q265 DIFFICULT Difficult/Tricky

A table has a primary key defined on a single column, but the DBMS is misconfigured so that duplicate values are silently accepted. Which fundamental rule is effectively broken, even though the column is still labeled 'primary key'?

A. Referential integrity

B. Domain integrity

C. Entity integrity

D. No rule is broken, since it is still labeled a primary key

Ans: C

Explanation

A primary key that permits duplicates fails the uniqueness requirement of entity integrity, regardless of how the column is labeled.

Q266 DIFFICULT Difficult/Tricky

Assertion (A): A foreign key column can safely be left without a NOT NULL constraint while still fully respecting referential integrity.
Reason (R): Referential integrity allows a foreign key value to be NULL, as long as it is not required to match a parent key.
Choose the correct option.

A. Both A and R are false

B. A is false but R is true

C. A is true but R is false

D. A is true and R is true, and R correctly explains A

Ans: D

Explanation

Referential integrity permits an optional relationship via a NULL foreign key, so leaving out NOT NULL does not itself break referential integrity; R correctly explains this.

Q267 DIFFICULT Difficult/Tricky

A table's primary key column has a UNIQUE constraint but is NOT explicitly marked NOT NULL. One row has a NULL value in this column. Is entity integrity satisfied?

A. Yes, since UNIQUE alone guarantees entity integrity

B. Cannot be determined without further information

C. No, because entity integrity also requires the primary key to be non-NULL, not merely unique

D. Yes, because NULL is treated as a valid unique value

Ans: C

Explanation

Entity integrity requires BOTH uniqueness AND a non-NULL primary key; a UNIQUE constraint alone does not guarantee the non-NULL requirement is met.

Q268 DIFFICULT Difficult/Tricky

A CHILD table's foreign key column allows NULL. A row is inserted with the foreign key set to NULL. Later, a report incorrectly claims this is a referential integrity violation. Why is the report WRONG?

A. Because NULL foreign keys always match every parent row

B. Because referential integrity explicitly permits a NULL foreign key value when the column allows NULL, so no violation has occurred

C. Because the report is actually correct in this case

D. Because referential integrity does not apply to NULL values under any interpretation

Ans: B

Explanation

A NULL foreign key, when permitted by the column definition, does not violate referential integrity — it simply represents the absence of a relationship.

Q269 DIFFICULT Difficult/Tricky

Which of the following is the BEST reason an assertion is sometimes preferred over multiple individual CHECK constraints spread across tables?

A. CHECK constraints cannot exist in a relational database

B. An assertion can express a single, unified condition across multiple tables, which would otherwise require complex coordination between separate per-table constraints

C. Assertions replace the need for primary keys

D. An assertion is always faster to execute than any CHECK constraint

Ans: B

Explanation

An assertion's strength lies in expressing one coherent cross-table condition, avoiding the need to coordinate multiple separate, table-bound CHECK constraints.

Q270 DIFFICULT Difficult/Tricky

A trigger is defined to fire AFTER an INSERT on the ORDERS table, and it inserts a row into the AUDIT table. If the INSERT into ORDERS itself fails (e.g., due to an entity integrity violation), what happens to the trigger?

A. The trigger does not fire, since the AFTER INSERT event only occurs upon a successful insert

B. The trigger still fires, since triggers are independent of the base operation's success

C. The trigger converts the failed insert into a successful one

D. The trigger fires twice to compensate

Ans: A

Explanation

An AFTER trigger only fires once its associated event (here, a successful INSERT) has actually completed; a failed insert due to an integrity violation means the triggering event never truly occurred.

Q271 DIFFICULT Difficult/Tricky

Which of the following correctly identifies the SUBTLE difference between a domain integrity violation and an entity integrity violation, when a primary key column also has a restricted data type?

A. A NULL or duplicate value in that column is an entity integrity issue, while a value of the wrong data type or out-of-range value (that is still unique and non-NULL) is a domain integrity issue

B. Domain integrity only applies to foreign keys, so this scenario is impossible

C. Entity integrity always overrides domain integrity, making the latter irrelevant

D. They are indistinguishable in every case

Ans: A

Explanation

Even on a primary key column, uniqueness/non-NULL violations fall under entity integrity, while value-validity issues (wrong type, out of range) fall under domain integrity — the two rules can apply to the same column but govern different aspects.

Q272 DIFFICULT Difficult/Tricky

Assertion (A): Referential integrity can be maintained purely through triggers, without a formal FOREIGN KEY constraint.
Reason (R): A well-designed trigger can check for a matching parent row before allowing an insert or update on the child table's foreign key column.
Choose the correct option.

A. A is true but R is unrelated to A

B. A is true and R is true, and R correctly explains how A could be achieved

C. A is false and R is false

D. A is false but R is true and unrelated

Ans: B

Explanation
While a formal FOREIGN KEY constraint is the standard approach, it is conceptually true that a carefully written trigger could check for a matching parent row and reject invalid operations, achieving a similar effect; R correctly explains this.

Q273 DIFFICULT Difficult/Tricky

ON DELETE SET NULL is defined, but the foreign key column is declared NOT NULL. A referenced parent row is deleted. What happens?

A. The delete succeeds and the child rows keep their old key

B. The child rows are deleted instead

C. The child keys become 0

D. The delete fails, because NULL cannot be stored in the column

Ans: D

Explanation
SET NULL needs a nullable column. With NOT NULL the operation cannot be completed.

Q274 DIFFICULT Difficult/Tricky

ON DELETE SET DEFAULT is used and the column default is 99, but no parent row has key 99. A parent row is deleted. What is the outcome?

A. 99 is added to the parent table

B. The operation fails, since the default would violate referential integrity

C. The children get NULL

D. The children are deleted

Ans: B

Explanation
The default value must match an existing parent key, otherwise the new reference is itself invalid.

Q275 | **DIFFICULT** | **Difficult/Tricky**

EMP (EmpID PK, ManagerID FK referring to EmpID) uses ON DELETE CASCADE. The top manager is deleted. What follows?

A. Only the direct reports are deleted

B. The deletion is rejected

C. Nothing else changes

D. All employees reporting to that manager, directly or indirectly, are deleted

Ans: D

Explanation

Cascading repeats down the chain: each deleted employee triggers deletion of the rows that refer to them.

Q276 | **MODERATE** | **Difficult/Tricky**

A designer makes Name the primary key of STUDENT. A second student called 'Amit' is enrolled. What results, and why?

A. The insert succeeds, because names may repeat

B. The insert fails with a duplicate key, because primary key values must be unique

C. The insert fails because of referential integrity

D. The insert succeeds, but the first Amit is removed

Ans: B

Explanation

Name is a poor key choice, since entity integrity forbids duplicate primary key values.

Q277 | **DIFFICULT** | **Difficult/Tricky**

DeptID in EMPLOYEE is named like DEPARTMENT's key, but no FOREIGN KEY is declared. A row with DeptID 99, absent from DEPARTMENT, is inserted. What happens?

A. It is rejected, because 99 is out of the domain

B. It is accepted, because referential integrity is not enforced without a declared foreign key

C. It is accepted, and 99 is added to DEPARTMENT

D. It is rejected, because the names match

Ans: B

Explanation

The DBMS enforces only declared constraints. Matching names alone create no reference.

Q278 DIFFICULT Difficult/Tricky

Consider the statements:
I. Inserting a row into the parent table can never violate referential integrity.
II. Deleting a row from the parent table can violate it.
III. Updating a foreign key in the child table can violate it.
Which statements are correct?

A. II and III only

B. II only

C. I, II and III

D. I and II only

Ans: C

Explanation
A new parent key is never referenced yet, so I holds. II and III describe the usual ways to create orphans.

Q279 DIFFICULT Difficult/Tricky

Assertion (A): A table can satisfy entity integrity and still violate domain integrity.
Reason (R): Entity integrity concerns only the uniqueness and non-NULL status of the primary key, not the validity of values.
Choose the correct option.

A. Both A and R are true, and R explains A

B. A is false but R is true

C. A is true but R is false

D. Both A and R are true, but R does not explain A

Ans: A

Explanation
A row can have a valid new key and yet hold an out-of-range value elsewhere, so the two rules are independent.

Q280 MODERATE Difficult/Tricky

RollNo is the primary key and Age must be between 15 and 30. The row (7, 45) is inserted, where 7 is a new key. Which statement is true?

A. Entity integrity is violated, domain integrity holds

B. Both are violated

C. Entity integrity holds, but domain integrity is violated

D. Neither is violated, since Age is not a key

Ans: C

Explanation
The key 7 is unique and non-NULL, but Age 45 is outside the allowed range.

SECTION 11

Mixed OAVS TGT Practice Set

Q281 | **EASY** | **Mixed Practice**

Which integrity rule ensures that a table's primary key is always unique and non-NULL?

A. Referential integrity

B. Assertion

C. Domain integrity

D. Entity integrity

Ans: D

Explanation

This is the defining requirement of entity integrity.

Q282 | **EASY** | **Mixed Practice**

A foreign key value that does not exist in the parent table's primary key causes a violation of:

A. Domain integrity

B. Entity integrity

C. No violation

D. Referential integrity

Ans: D

Explanation

This is the classic definition of a referential integrity violation.

Q283 | **MODERATE** | **Mixed Practice**

Which mechanism automatically executes a defined set of actions when a specified data event occurs?

A. Assertion

B. Entity constraint

C. Trigger

D. Domain constraint

Ans: C

Explanation

A trigger is defined precisely for automatic, event-driven execution of actions.

Q284 | **MODERATE** | **Mixed Practice**

Which mechanism is most suitable for enforcing a condition that spans multiple tables and must always remain true?

A. A simple NOT NULL constraint

B. Entity integrity

C. An assertion

D. Domain integrity

Ans: C

Explanation

An assertion is designed to express and enforce broad, multi-table conditions that must always hold true.

Q285 | **MODERATE** | **Mixed Practice**

A column 'Marks' must lie between 0 and 100. Which type of integrity governs this?

A. Trigger-based integrity

B. Entity integrity

C. Referential integrity

D. Domain integrity

Ans: D

Explanation

Range restriction on a single column's values is a domain integrity rule.

Q286 | **EASY** | **Mixed Practice**

Which of these best describes the ECA concept associated with triggers?

A. Entity-Check-Assertion

B. Error-Correction-Analysis

C. Entity-Constraint-Assertion

D. Event-Condition-Action

Ans: D

Explanation

ECA stands for Event-Condition-Action, describing how a trigger fires.

Q287 | MODERATE | Mixed Practice

STUDENT(RollNo PK) has a row with RollNo = NULL. Which rule is violated?

A. Referential integrity

B. No violation

C. Entity integrity

D. Domain integrity

Ans: C

Explanation

A NULL primary key value violates entity integrity.

Q288 | MODERATE | Mixed Practice

EMPLOYEE(DeptID FK) has DeptID = 55, which does not exist in DEPARTMENT(DeptID PK). Which rule is violated?

A. Entity integrity

B. Domain integrity

C. Referential integrity

D. Assertion

Ans: C

Explanation

An unmatched foreign key value violates referential integrity.

Q289 | MODERATE | Mixed Practice

Which of the following is TRUE about NULL values and the different integrity rules?

A. A primary key cannot be NULL (entity integrity), but a foreign key may be NULL if the column allows it (referential integrity)

B. Both primary keys and foreign keys can always be NULL

C. NULL is never allowed anywhere in any table

D. NULL is only relevant to domain integrity

Ans: A

Explanation

This correctly distinguishes the NULL rules: strictly disallowed for primary keys (entity integrity), but permitted for foreign keys where the column allows it (referential integrity).

Q290 | **MODERATE** | **Mixed Practice**

Which of the following would be enforced using a CHECK-type constraint, conceptually tied to domain integrity?

A. A rule that DeptID must exist in DEPARTMENT

B. A rule that fires an email whenever a new employee is added

C. A rule that Salary must be greater than 0

D. A rule that EmpID must be unique

Ans: C

Explanation

Restricting a column's values to a valid range (Salary > 0) is a domain integrity rule, typically implemented via a CHECK constraint.

Q291 | **DIFFICULT** | **Mixed Practice**

Which single statement best summarizes the difference between the three basic integrity rules (entity, domain, referential) and the two advanced mechanisms (assertion, trigger)?

A. There is no difference; all five are the same

B. The three basic rules address fixed, well-known patterns of key uniqueness, valid values, and valid references; assertions and triggers extend integrity enforcement to custom, database-wide, or event-driven conditions

C. Assertions and triggers replace the need for entity, domain, and referential integrity

D. Only entity integrity is enforced automatically by any DBMS

Ans: B

Explanation

This captures the syllabus's core theme: entity, domain, and referential integrity cover standard, predictable rules, while assertions and triggers provide flexible, general-purpose or automatic enforcement beyond those fixed patterns.

Q292 | **MODERATE** | **Mixed Practice**

Consider the statements:
I. A foreign key may be NULL.
II. A primary key may be NULL.
III. A trigger fires automatically on its event.
IV. An assertion is activated by a single event.
Which statements are correct?

A. I, II and III

B. II and III only

C. I and III only

D. I and IV only

Ans: C

Explanation
Primary keys cannot be NULL and an assertion is a standing condition, not an event-driven action.

Q293 | **EASY** | **Mixed Practice**

In what order does a trigger work?

A. Condition, then action, then event

B. Event, then action, then condition

C. Event, then condition, then action

D. Action, then event, then condition

Ans: C

Explanation
This is the Event-Condition-Action model.

Q294 | **MODERATE** | **Mixed Practice**

Fill in the blanks: Domain integrity concerns _____, entity integrity concerns _____, and referential integrity concerns _____.

A. attribute values; primary keys; links between tables

B. primary keys; attribute values; links between tables

C. attribute values; links between tables; primary keys

D. links between tables; primary keys; attribute values

Ans: A

Explanation
Each rule has its own focus: values, the primary key, and key relationships between tables.

Q295 | **MODERATE** | **Mixed Practice**

Which action can a trigger perform that a plain constraint cannot?

A. Automatically writing a log row when a record is deleted

B. Rejecting a value outside a range

C. Rejecting a duplicate primary key

D. Rejecting an unmatched foreign key

Ans: A

Explanation

Constraints accept or reject; only a trigger can carry out an extra action such as logging.

Q296 | **EASY** | **Mixed Practice**

Which mechanism prevents orphan rows in a child table?

A. Domain integrity

B. An assertion on a single column

C. Entity integrity

D. Referential integrity

Ans: D

Explanation

Orphans are child rows with no matching parent, which referential integrity forbids.

Q297 | **MODERATE** | **Mixed Practice**

A table has primary key EmpID, foreign key DeptID, a NOT NULL column Name and a plain column Phone. Which columns may hold NULL, subject to their definitions?

A. DeptID and Phone

B. Name and Phone

C. EmpID only

D. All four columns

Ans: A

Explanation

A primary key and a NOT NULL column cannot be NULL. A foreign key and an ordinary column may be.

Q298 | **MODERATE** | **Mixed Practice**

The row (RollNo = NULL, Marks = 120, DeptID = 'D9', not present in DEPT) is inserted. Which rules are violated?

A. Domain and referential integrity only

B. Entity integrity only

C. Entity, domain and referential integrity

D. Entity and referential integrity only

Ans: C

Explanation

NULL breaks the primary key rule, 120 is out of the marks range, and D9 has no parent.

Q299 | **EASY** | **Mixed Practice**

Which property do entity, domain and referential integrity share?

A. All perform user-defined actions on events

B. All keep data valid and consistent through rules the DBMS enforces

C. All apply only to foreign keys

D. All are checked only when a table is created

Ans: B

Explanation

They are integrity constraints enforced automatically by the DBMS.

Q300 | **EASY** | **Mixed Practice**

Which integrity rule is checked at the level of a single attribute's values?

A. Entity integrity

B. Referential integrity

C. Domain integrity

D. Assertion

Ans: C

Explanation

Domain integrity restricts the values a single attribute may hold.